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Primer-BLAST

» **JOB ID:bWewTvIT9PvTxfHA_KDV8oa7xMCrqN_dqg**

Primer-BLAST Results

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•

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Input PCR template

none

Specificity of primers

Target templates were found in selected database: Core nucleotide BLAST database (Organism limited to Hypocreales)

Other reports

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Detailed primer reports + −

You can re-search for specific primers by accepting some of the unintended targets, check the box(es) next to the ones you accept and try again to re-search for specific primers Submit

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- Download primer pairs ▼
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Primer pair 1

	Sequence (5'->3')	Length	Tm	GC%	Self complementarity	Self 3' complementarity
Forward primer	TGGTACAAGGGNTGGGAGAAGG	22	58.56	57.14	4.00	0.00
Reverse primer	GCTGCTCGTGGTGCATNTCN	20	56.47	61.11	4.00	1.00

Products on intended targets

Products on allowed targets

Products on allowed transcript variants

Products on potentially unintended templates

Products on target templates

>[KJ536074.1](#) Beauveria bassiana strain EABb 10/84-Fil translation elongation factor 1-alpha (tef1a) gene, partial cds

```
product length = 268
Forward primer  1  TGGTACAAGGGNTGGGAGAAGG  22
Template        224  .....C.....  245

Reverse primer  1  GCTGCTCGTGGTGCATNTCN  20
Template        491  .....C..G  472
```

>[PV855311.1](#) Clonostachys rosea isolate DX-2-3B translation elongation factor 1-alpha (tef1) gene, partial cds

```
product length = 268
Forward primer  1  TGGTACAAGGGNTGGGAGAAGG  22
Template        295  .....T.....  316
```

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 562C..A 543

>[DQ069992.1](#) *Aschersonia aleyrodis* strain CUP67342 elongation factor 1-alpha gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 186T..... 207

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 453T..G 434

>[OP963749.1](#) *Beauveria australis* strain DAR86168 translation elongation factor 1-alpha (EF1a) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 308T..... 329

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 575C..G 556

>[KP009011.1](#) *Trichoderma asperellum* strain TRS705 translation elongation factor 1-alpha (tef1) gene, exons 5 and 6 and partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 829C..... 850

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1096C..G 1077

>[OQ440677.1](#) *Beauveria* sp. voucher KUMCC 8636 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 285T..... 306

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 552C..G 533

>[JX083689.1](#) *Claviceps microcephala* isolate CCC1094 translation elongation factor 1-alpha (EF1-alpha) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 325C..... 346

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 592T..C 573

>[KJ665690.1](#) *Trichoderma priscilae* strain S144 translation elongation factor 1-alpha (tef1) gene, partial cds

```

product length = 268
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template        819    .....C.....  840

Reverse primer  1      GCTGCTCGTGGTGCATNTCN  20
Template        1086   .....C..G  1067

```

>[ON934358.1](#) *Trichoderma orientale* isolate E225 translation elongation factor 1-alpha (TEF1) gene, partial cds

```

product length = 268
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template        802    .....C.....  823

Reverse primer  1      GCTGCTCGTGGTGCATNTCN  20
Template        1069   .....C..A  1050

```

>[MN523563.1](#) *Beauveria pseudobassiana* voucher YFCC 1806012 translation elongation factor 1-alpha (TEF) gene, partial cds

```

product length = 268
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template        328    .....T.....  349

Reverse primer  1      GCTGCTCGTGGTGCATNTCN  20
Template        595    .....C..G  576

```

>[MT430097.1](#) *Claviceps purpurea* isolate LM567 translation elongation factor 1-alpha (TEF1-alpha) gene, partial cds

```

product length = 268
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template        208    .....C.....  229

Reverse primer  1      GCTGCTCGTGGTGCATNTCN  20
Template        475    .....T..C  456

```

>[LC370945.1](#) *Ophiocordyceps* sp. gene for translation elongation factor 1-alpha, partial cds, strain: MosKNG01_OP

```

product length = 268
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template        235    .....C.....  256

Reverse primer  1      GCTGCTCGTGGTGCATNTCN  20
Template        502    .....C..G  483

```

>[PV273549.1](#) *Emericellopsis terricola* strain CBS 187.68 translation elongation factor 1-alpha (TEF1) gene, partial cds

```

product length = 268
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template        132    .....C.....  153

Reverse primer  1      GCTGCTCGTGGTGCATNTCN  20
Template        399    .....T..A  380

```

>[MW267411.1](#) *Trichoderma harzianum* isolate TA2-46 translation elongation factor 1-alpha (TEF-1 alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 785C..... 806

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1052C..G 1033

>[OR066640.1](#) *Ophiocordyceps* sp. isolate CMAFB01 translation elongation factor 1-alpha (EF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 235C..... 256

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 502C..G 483

>[MT430099.1](#) *Claviceps purpurea* isolate LM565 translation elongation factor 1-alpha (TEF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 208C..... 229

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 475T..C 456

>[EU369023.1](#) *Torrubiella patchii* strain NHJ 6209 translation elongation factor 1 alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 328C..... 349

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 595C..G 576

>[MW292447.1](#) *Ophiocordyceps asiana* isolate MY11880 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 232T..... 253

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 499T..G 480

>[PQ658983.1](#) *Beauveria bassiana* strain NR-OM27 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 232T..... 253

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 499C..G 480

>[GU233716.1](#) *Beauveria bassiana* strain RCEF0760 translation elongation factor 1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 54T..... 75

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 321C..G 302

>[LC598980.1](#) *Claviceps* sp. TNS:F60482 Ef1-alpha gene for translation elongation factor 1-alpha, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 326C..... 347

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 593T..C 574

>[JX083688.1](#) *Claviceps microcephala* isolate CCC1031 translation elongation factor 1-alpha (EF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 325C..... 346

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 592T..C 573

>[OQ980388.1](#) *Emericellopsis fuci* voucher TR:061aS1.2 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 241C..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 508C..G 489

>[OP620752.1](#) *Trichoderma tomentosum* isolate XY101314 translation elongation factor 1-alpha (EF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 289C..... 310

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 556C..G 537

>[MT215058.1](#) *Beauveria bassiana* isolate BG42 translation elongation factor 1-alpha gene, partial cds

```

product length = 268
Forward primer 1   TGGTACAAGGGNTGGGAGAAGG  22
Template       249 .....T.....  270

Reverse primer 1   GCTGCTCGTGGTGCATNTCN  20
Template       516 .....C..G  497

```

>[MH795106.1](#) *Sphaerostilbella himalayensis* voucher TFC:201711 elongation factor 1-alpha (EF1-alpha) gene, partial cds

```

product length = 268
Forward primer 1   TGGTACAAGGGNTGGGAGAAGG  22
Template       744 .....C.....  765

Reverse primer 1   GCTGCTCGTGGTGCATNTCN  20
Template      1011 .....C..A  992

```

>[AF534613.1](#) *Trichoderma stromaticum* strain PC 209 translation elongation factor 1-alpha (ef1-alpha) gene, partial cds

```

product length = 268
Forward primer 1   TGGTACAAGGGNTGGGAGAAGG  22
Template       183 .....C.....  204

Reverse primer 1   GCTGCTCGTGGTGCATNTCN  20
Template      450 .....C..G  431

```

>[OP620747.1](#) *Trichoderma pollinicola* isolate PL21811-12-2 translation elongation factor 1-alpha (EF1-alpha) gene, partial cds

```

product length = 268
Forward primer 1   TGGTACAAGGGNTGGGAGAAGG  22
Template       260 .....C.....  281

Reverse primer 1   GCTGCTCGTGGTGCATNTCN  20
Template      527 .....C..G  508

```

>[PQ433542.1](#) *Cosmospora* sp. strain 1834 translation elongation factor 1-alpha gene, partial cds

```

product length = 268
Forward primer 1   TGGTACAAGGGNTGGGAGAAGG  22
Template       241 .....C.....  262

Reverse primer 1   GCTGCTCGTGGTGCATNTCN  20
Template      508 .....C..G  489

```

>[OQ471097.1](#) *Paracylindrocarpon multiseptatum* isolate CBS336.77 translation elongation factor 1-alpha (tef1) gene, partial cds

```

product length = 268
Forward primer 1   TGGTACAAGGGNTGGGAGAAGG  22
Template       241 .....C.....  262

Reverse primer 1   GCTGCTCGTGGTGCATNTCN  20
Template      508 .....C..G  489

```

>[OQ809062.1](#) *Sedecimiella funiculosus* strain CGMCC 3.22348 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 286C..... 307

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 553C..A 534

>[OL440152.1](#) *Thyronectria ulmi* strain TCH1 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 833A..... 854

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1100C..G 1081

>[LC163556.1](#) *Beauveria bassiana* EF1-a gene for translation elongation factor 1 alpha, partial cds, isolate: F932

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 326T..... 347

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 593C..G 574

>[KP008834.1](#) *Trichoderma aff. harzianum* MO-2014 strain TRS597 translation elongation factor 1-alpha (tef1) gene, exons 5 and 6 and partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 743C..... 764

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1010C..G 991

>[EF468769.1](#) *Cordyceps* sp. EFCC 2135 translation elongation factor 1 alpha-like (tef) gene, partial sequence

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 328C..... 349

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 595T..C 576

>[OQ440720.1](#) *Purpureocillium lilacinum* voucher KUMCC 9352 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 285C..... 306

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 552C..A 533

>[PV153472.1](#) *Moelleriella* sp. voucher LF202207001C translation elongation factor 1-alpha (tef1) gene, partial cds; mitochondrial

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 297A..... 318

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 564T..G 545

>[MW267408.1](#) *Trichoderma harzianum* isolate TA2-38 translation elongation factor 1-alpha (TEF-1 alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 786C..... 807

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1053C..G 1034

>[PV617177.1](#) *Trichoderma* sp. strain CFCC 51933 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 716C..... 737

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 983C..G 964

>[LC572040.1](#) *Aciculosporium oplismeni* MAFF:246966 tef1a gene for translation elongation factor 1 alpha, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 325C..... 346

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 592T..A 573

>[EF193190.1](#) *Beauveria bassiana* isolate KVL 03_85 translation elongation factor 1 alpha gene, partial sequence

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 1055T..... 1076

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1322C..G 1303

>[MF374810.1](#) *Trichoderma zonatum* strain 225 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 843C..... 864

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1110C..G 1091

>[ON326405.1](#) *Hevansia* sp. TW-2022a voucher NL20210822-05 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 295T..... 316

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 562C..G 543

>[GU979994.1](#) *Isaria takamizusanensis* strain F896 translation elongation factor 1 alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 297C..... 318

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 564C..G 545

>[OQ470981.1](#) *Lasionectria antillana* isolate CBS122797 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 241T..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 508C..G 489

>[OR548086.1](#) *Trichoderma thailandicum* isolate 1283 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 819C..... 840

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1086C..G 1067

>[OR667292.1](#) *Trichoderma harzianum* culture SZMC:0560 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 815C..... 836

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1082C..G 1063

>[PV273474.1](#) *Stilbocrea* sp. strain CBS 659.83 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 241C..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 508C..C 489

>[LC769719.1](#) *Beauveria bassiana* BCRC:FU31676 tef gene for translation elongation factor 1-alpha, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 309T..... 330

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 576C..G 557

>[MN576985.1](#) *Cordyceps cicadae* strain YFCC 7128 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 316T..... 337

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 583C..G 564

>[OQ471234.1](#) *Valsonectria simpsonii* isolate CBS101602 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 274

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 247T..... 268

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 520C..A 501

>[PV755856.1](#) *Trichoderma* sp. strain CFCC 85490A translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 818C..... 839

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1085C..A 1066

>[OR115229.1](#) *Papiliomyces shibinensis* strain YFCC 859.01 translation elongation factor 1 alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 328T..... 349

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 595T..C 576

>[PP965294.1](#) *Gibellula* sp. strain C13 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 197C..... 218

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 464T..G 445

>[OQ858798.1](#) *Trichoderma aff. hortense* AD46-S3-2 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 812C..... 833

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1079C..G 1060

>[OR842382.1](#) *Moelleriella* sp. ZY-2023d strain YFCC 22099303 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 309T..... 330

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 576C..G 557

>[ON512611.1](#) *Tolypocladium* sp. isolate JMS370 translation elongation factor EF1-alpha (tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 244C..... 265

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 511C..C 492

>[KJ570761.1](#) *Thyronectria austroamericana* strain GG translation elongation factor 1 alpha (TEF1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 805A..... 826

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1072C..G 1053

>[PV273536.1](#) *Emericellopsis mexicana* strain CBS 125295 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 241C..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 508C..A 489

>[PV617263.1](#) *Trichoderma* sp. strain CFCC 88662 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 708T..... 729

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 975C..G 956

>[MG595717.1](#) *Trichoderma asperellum* isolate T2007 translation elongation factor 1-alpha (tef1alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 821C..... 842

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1088C..G 1069

>[OM482381.1](#) *Cordyceps fumosorosea* strain RCEF2540 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 295C..... 316

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 562C..A 543

>[PV273537.1](#) *Emericellopsis mexicana* strain CBS 179.92 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 241C..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 508C..A 489

>[MW267387.1](#) *Trichoderma harzianum* isolate TRB2-2 translation elongation factor 1-alpha (TEF-1 alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 789C..... 810

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1056C..G 1037

>[KP689567.1](#) *Epichloe mollis* strain AL9924 translation elongation factor 1-alpha (tefA) gene, complete cds

```
product length = 268
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template        1113  .....T.....  1134

Reverse primer  1      GCTGCTCGTGGTGCATNTCN  20
Template        1380  .....T..A  1361
```

>[PX225635.1](#) *Ophiocordyceps* sp. YG-2025c voucher GMB 3129 translation elongation factor 1-alpha gene, partial cds

```
product length = 268
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template        331  .....C.....  352

Reverse primer  1      GCTGCTCGTGGTGCATNTCN  20
Template        598  .....T..A  579
```

>[PP982895.1](#) *Hypomyces* sp. isolate MFLUCC 24-0218 translation elongation factor alpha-1 (tef1) gene, partial cds

```
product length = 268
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template        292  .....C.....  313

Reverse primer  1      GCTGCTCGTGGTGCATNTCN  20
Template        559  .....C..G  540
```

>[KM411987.1](#) *Ophiocordyceps amazonica* isolate Ophama1856 translation elongation factor 1-alpha (TEF) gene, partial cds

```
product length = 268
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template        226  .....C.....  247

Reverse primer  1      GCTGCTCGTGGTGCATNTCN  20
Template        493  .....C..C  474
```

>[LC510261.1](#) *Isaria macroscyticola* 170628-1 tef1 gene for translation elongation factor 1-alpha, partial cds

```
product length = 268
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template        242  .....C.....  263

Reverse primer  1      GCTGCTCGTGGTGCATNTCN  20
Template        509  .....C..A  490
```

>[KY973654.1](#) *Cordyceps tenuipes* voucher ARSEF 5135 elongation factor-1 alpha gene, partial cds

```
product length = 268
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template        247  .....T.....  268

Reverse primer  1      GCTGCTCGTGGTGCATNTCN  20
```

Template 514C..A 495

>[MW292449.1](#) *Ophiocordyceps asiana* isolate MY11884 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 283T..... 304

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 550T..G 531

>[KJ636534.1](#) *Pochonia* sp. LS-2013b isolate AR5115-2 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 310T..... 331

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 577C..C 558

>[PV755809.1](#) *Trichoderma* sp. strain CFCC 53255A translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 796C..... 817

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1063C..G 1044

>[LT216543.1](#) *Claviceps viridis* partial tef1a gene for translation elongation factor 1 alpha, strain CBS 125.63

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 325T..... 346

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 592T..C 573

>[KJ665571.1](#) *Trichoderma mediterraneum* strain S240 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 829T..... 850

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1096C..G 1077

>[MN433214.1](#) *Nectria marina* voucher MFLUCC 16-0544 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 340C..... 361

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 607C..G 588

>[KJ878956.1](#) *Ophiocordyceps formosana* voucher TNM F13893 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 281C..... 302

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 548C..A 529

>[MH203479.1](#) *Beauveria bassiana* strain CHE-CNRCB 160 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 340T..... 361

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 607C..G 588

>[MT429996.1](#) *Claviceps purpurea* isolate LM230 translation elongation factor 1-alpha (TEF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 208C..... 229

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 475T..C 456

>[MT881861.1](#) *Trichoderma longibrachiatum* strain 18ELIB014 translation initiation factor 1 alpha (TEF1a) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 828C..... 849

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1095C..A 1076

>[EU248857.1](#) *Metarhizium guizhouense* culture-collection ARSEF:6238 translation elongation factor 1 alpha (TEF) gene, partial sequence

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 1032T..... 1053

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1299T..C 1280

>[KC870070.1](#) *Metarhizium pinghaense* isolate RMT29 translation elongation factor 1-alpha (EF1alpha) gene, partial cds

```
product length = 268
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template       994      .....T.....  1015

Reverse primer  1      GCTGCTCGTGGTGCATNTCN  20
Template       1261     .....T..C  1242
```

>[MH795102.1](#) *Sphaerostilbella toxica* voucher TFC:202258 elongation factor 1-alpha (EF1-alpha) gene, partial cds

```
product length = 268
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template       811      .....T.....  832

Reverse primer  1      GCTGCTCGTGGTGCATNTCN  20
Template       1078     .....C..A  1059
```

>[KX632600.1](#) *Trichoderma harzianum* strain T11 translation elongation factor 1-alpha (tef1) gene, partial cds

```
product length = 268
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template       797      .....C.....  818

Reverse primer  1      GCTGCTCGTGGTGCATNTCN  20
Template       1064     .....C..G  1045
```

>[LC684217.1](#) *Claviceps paspali* TNS:F-96506 TEF gene for translation elongation factor 1-alpha, partial sequence

```
product length = 268
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template       326      .....C.....  347

Reverse primer  1      GCTGCTCGTGGTGCATNTCN  20
Template       593      .....T..C  574
```

>[MF939620.1](#) *Trichoderma pollinicola* strain LC11686 translation elongation factor 1-alpha (tef1) gene, partial cds

```
product length = 268
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template       795      .....C.....  816

Reverse primer  1      GCTGCTCGTGGTGCATNTCN  20
Template       1062     .....C..G  1043
```

>[OU720054.1](#) *Claviceps* sp. KP-2018a partial tef1a gene for translation elongation factor 1 alpha, strain CCC1427

```
product length = 268
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template       325      .....T.....  346
```

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 592T..C 573

>[OQ116869.1](#) *Trichoderma simmonsii* isolate T57.0.1 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 798C..... 819

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1065C..G 1046

>[PV755789.1](#) *Trichoderma* sp. strain CFCC 51121A translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 786C..... 807

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1053C..A 1034

>[MN605880.1](#) *Trichoderma lentinulae* strain CGMCC 3.19849 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 807C..... 828

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1074C..G 1055

>[GU979977.1](#) *Paecilomyces lilacinus* strain MY683 translation elongation factor 1 alpha-like (tef) gene, partial sequence

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 339C..... 360

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 606C..A 587

>[MW723128.1](#) *Metarhizium viridulum* strain RCEF6807 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 296T..... 317

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 563C..C 544

>[LC496913.1](#) *Simplicillium* sp. JCM 39050 TEF1a gene for translation elongation factor 1 alpha, partial sequence

```

product length = 268
Forward primer  1    TGGTACAAGGGNTGGGAGAAGG  22
Template        298  .....A.....  319

Reverse primer  1    GCTGCTCGTGGTGCATNTCN  20
Template        565  .....C..G  546

```

>[MH244849.1](#) *Cordyceps fumosorosea* strain AMEF067 translation elongation factor 1-alpha gene, partial cds

```

product length = 268
Forward primer  1    TGGTACAAGGGNTGGGAGAAGG  22
Template        252  .....C.....  273

Reverse primer  1    GCTGCTCGTGGTGCATNTCN  20
Template        519  .....C..A  500

```

>[MT430046.1](#) *Claviceps purpurea* isolate LM1025 translation elongation factor 1-alpha (TEF1-alpha) gene, partial cds

```

product length = 268
Forward primer  1    TGGTACAAGGGNTGGGAGAAGG  22
Template        208  .....C.....  229

Reverse primer  1    GCTGCTCGTGGTGCATNTCN  20
Template        475  .....T..C  456

```

>[OR515611.1](#) *Metarhizium anisopliae* isolate 10005-1 translation elongation factor 1-alpha (EF1alpha) gene, partial cds

```

product length = 268
Forward primer  1    TGGTACAAGGGNTGGGAGAAGG  22
Template        302  .....T.....  323

Reverse primer  1    GCTGCTCGTGGTGCATNTCN  20
Template        569  .....T..C  550

```

>[OQ541844.1](#) *Trichoderma citrinoviride* strain MRB1901-232 translation elongation factor 1-alpha (tef1) gene, partial cds

```

product length = 268
Forward primer  1    TGGTACAAGGGNTGGGAGAAGG  22
Template        859  .....C.....  880

Reverse primer  1    GCTGCTCGTGGTGCATNTCN  20
Template        1126  .....C..G  1107

```

>[OR296642.1](#) *Metarhizium anisopliae* strain ARSEF 798 translation elongation factor 1-alpha (EF1a) gene, partial cds

```

product length = 268
Forward primer  1    TGGTACAAGGGNTGGGAGAAGG  22
Template        277  .....T.....  298

Reverse primer  1    GCTGCTCGTGGTGCATNTCN  20
Template        544  .....T..C  525

```

>[KJ665715.1](#) *Trichoderma simmonsii* strain S355 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 810C..... 831

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1077C..G 1058

>[PQ369657.1](#) *Ophiocordyceps* sp. WB-2024b voucher HKAS 126122 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 295T..... 316

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 562T..G 543

>[MT671925.1](#) *Trichoderma longibrachiatum* strain 18ASMA005 translation elongation factor 1-alpha (TEF1a) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 829C..... 850

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1096C..A 1077

>[MN733824.1](#) *Hirsutella hongheensis* voucher HKAS102451 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 158C..... 179

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 425C..G 406

>[MF362085.1](#) *Cordyceps takaomontana* voucher ANDES-F 1286 elongation factor-1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 236T..... 257

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 503C..A 484

>[KM652019.1](#) *Hirsutella thompsonii* voucher ARSEF 257 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 295C..... 316

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 562C..G 543

>[MH203493.1](#) *Beauveria bassiana* strain CHE-CNRCB 491 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 299T..... 320

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 566C..G 547

>[MK850824.1](#) *Trichoderma atroviride* isolate HZA2 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 826C..... 847

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1093C..A 1074

>[MH724259.1](#) *Escovopsis kreiselii* strain LESF302 translation elongation factor 1-alpha (tef1a) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 283C..... 304

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 550T..G 531

>[PV855316.1](#) *Clonostachys rosea* isolate MHK-1-6 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 295T..... 316

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 562C..A 543

>[PP982511.1](#) *Metarhizium rileyi* voucher CHE-CNRCB 682 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 1057C..... 1078

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1324T..C 1305

>[PV617237.1](#) *Trichoderma* sp. strain CFCC 81909 translation elongation factor 1-alpha (tef1) gene, partial cds

```

product length = 268
Forward primer 1   TGGTACAAGGGNTGGGAGAAGG  22
Template       708 .....C.....  729

Reverse primer 1   GCTGCTCGTGGTGCATNTCN  20
Template       975 .....C..G  956

```

>[OQ980390.1](#) *Nectriella dacrymycella* culture CBS:139243 strain CLL 14055 translation elongation factor 1-alpha (tef1) gene, partial cds

```

product length = 268
Forward primer 1   TGGTACAAGGGNTGGGAGAAGG  22
Template       241 .....C.....  262

Reverse primer 1   GCTGCTCGTGGTGCATNTCN  20
Template       508 .....C..G  489

```

>[KT748535.1](#) *Beauveria bassiana* isolate ArgB21 translation elongation factor 1-alpha (TEF1a) gene, partial cds

```

product length = 268
Forward primer 1   TGGTACAAGGGNTGGGAGAAGG  22
Template       323 .....C.....  344

Reverse primer 1   GCTGCTCGTGGTGCATNTCN  20
Template       590 .....C..G  571

```

>[MT430023.1](#) *Claviceps purpurea* isolate LM30 translation elongation factor 1-alpha (TEF1-alpha) gene, partial cds

```

product length = 268
Forward primer 1   TGGTACAAGGGNTGGGAGAAGG  22
Template       208 .....C.....  229

Reverse primer 1   GCTGCTCGTGGTGCATNTCN  20
Template       475 .....T..C  456

```

>[KM652004.1](#) *Hirsutella rhossiliensis* voucher ARSEF 2931 translation elongation factor 1-alpha gene, partial cds

```

product length = 268
Forward primer 1   TGGTACAAGGGNTGGGAGAAGG  22
Template       302 .....C.....  323

Reverse primer 1   GCTGCTCGTGGTGCATNTCN  20
Template       569 .....C..A  550

```

>[PV670063.1](#) *Metarhizium anisopliae* isolate IIHRMa4 translation elongation factor 1-alpha (TEF1a) gene, partial cds

```

product length = 268
Forward primer 1   TGGTACAAGGGNTGGGAGAAGG  22
Template       199 .....T.....  220

Reverse primer 1   GCTGCTCGTGGTGCATNTCN  20
Template       466 .....T..C  447

```

>[EF468798.1](#) *Simplicillium obclavatum* strain CBS 311.74 translation elongation factor 1 alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 328A..... 349

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 595C..G 576

>[OL473530.1](#) *Cordyceps simaoensis* strain YFCC 8407 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 328T..... 349

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 595C..A 576

>[OQ506282.1](#) *Akanthomyces araneogenus* strain YFCC 2206935 translation elongation factor 1-alpha (TEF1a) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 328C..... 349

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 595C..G 576

>[OQ471069.1](#) *Niesslia mucida* isolate CBS404.66 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 241T..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 508C..G 489

>[OQ944544.1](#) *Clonostachys apocyni* culture CBS:956.73 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 241T..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 508C..A 489

>[OR831943.1](#) *Moelleriella* sp. ZY-2023g strain YHH 2209002 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 295T..... 316

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 562T..G 543

>[OM482392.1](#) *Purpureocillium atypicola* strain RCEF1887 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 295C..... 316

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 562C..A 543

>[OR098431.1](#) *Ophiocordyceps tortuosa* voucher YHH 2210003 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 295C..... 316

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 562C..A 543

>[KM510158.1](#) *Metarhizium guizhouense* strain RCEF4413 translation elongation factor 1 alpha (TEF) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 287T..... 308

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 554T..C 535

>[KM283818.1](#) *Cordyceps confragosa* strain CBS 102067 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 289T..... 310

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 556C..G 537

>[MN401480.1](#) *Beauveria asiatica* strain BCC1665 translation elongation factor 1 alpha (tef) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 283T..... 304

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 550C..G 531

>[OQ954485.1](#) *Amphichorda littoralis* strain FMR 20067 translation elongation factor 1-alpha (Tef1a) gene, partial cds

```

product length = 268
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template        289      .....C.....  310

Reverse primer  1      GCTGCTCGTGGTGCATNTCN  20
Template        556      .....C..A  537

```

>[KY688043.1](#) *Trichoderma longisporum* voucher HMAS:248843 translation elongation factor 1-alpha (TEF1) gene, partial cds

```

product length = 268
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template        820      .....C.....  841

Reverse primer  1      GCTGCTCGTGGTGCATNTCN  20
Template       1087      .....C..G  1068

```

>[MT430105.1](#) *Claviceps purpurea* isolate LM558 translation elongation factor 1-alpha (TEF1-alpha) gene, partial cds

```

product length = 268
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template        208      .....C.....  229

Reverse primer  1      GCTGCTCGTGGTGCATNTCN  20
Template        475      .....T..C  456

```

>[AY391993.1](#) *Trichoderma sinuosum* strain G.J.S. 95-203 translation elongation factor 1-alpha gene, partial cds

```

product length = 268
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template        183      .....C.....  204

Reverse primer  1      GCTGCTCGTGGTGCATNTCN  20
Template        450      .....C..G  431

```

>[MW723124.1](#) *Metarhizium viridulum* strain RCEF6803 translation elongation factor 1-alpha (TEF) gene, partial cds

```

product length = 268
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template        291      .....T.....  312

Reverse primer  1      GCTGCTCGTGGTGCATNTCN  20
Template        558      .....C..C  539

```

>[KJ665529.1](#) *Trichoderma koningii* strain S267 translation elongation factor 1-alpha (tef1) gene, partial cds

```

product length = 268
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template        816      .....C.....  837

Reverse primer  1      GCTGCTCGTGGTGCATNTCN  20
Template       1083      .....C..G  1064

```

>[KT224463.1](#) *Trichoderma* sp. 3 WTQ-2015 strain 10035 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 848C..... 869

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1115C..G 1096

>[MH062189.1](#) *Akanthomyces dipterigenus* culture CEP:155 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 283T..... 304

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 550C..G 531

>[PV755819.1](#) *Trichoderma* sp. strain CFCC 53424A translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 800C..... 821

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1067C..G 1048

>[PQ790165.1](#) *Trichoderma* sp. 2 MC-2025 strain DSF 004-004-002 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 335C..... 356

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 602C..G 583

>[OQ858763.1](#) *Trichoderma virens* isolate IV37-S3-3 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 798C..... 819

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1065C..G 1046

>[MF362094.1](#) *Cordyceps takaomontana* voucher ANDES-F 1310 elongation factor-1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 281T..... 302

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 548C..A 529

>[OK040704.1](#) *Gibellula trimorpha* strain BCC36526 translation elongation factor 1-alpha gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 229C..... 250

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 496C..G 477

>[MH795098.1](#) *Sphaerostilbella appalachiensis* voucher TU:119635 elongation factor 1-alpha (EF1-alpha) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 298M..... 319

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 565C..A 546

>[EU248848.1](#) *Metarhizium robertsii* culture-collection ARSEF:4739 translation elongation factor 1 alpha (TEF) gene, partial sequence

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 1029T..... 1050

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1296T..C 1277

>[PV107657.1](#) *Tolypocladium* sp. isolate DG999 translation elongation factor 1-alpha (TEF1-alpha) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 163C..... 184

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 430C..G 411

>[ON101303.1](#) *Aciculosporium take* isolate SICAUCC 22-0025 translation elongation factor 1-alpha (tefA) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 293C..... 314

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 560T..A 541

>[MT430200.1](#) *Claviceps purpurea* isolate Clav09x1 translation elongation factor 1-alpha (TEF1-

alpha) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 208C..... 229
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 475T..C 456

>[KJ665434.1](#) *Trichoderma balearicum* culture CBS:133222 strain S402 voucher WU:33371 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 887C..... 908
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1154C..G 1135

>[OU720071.1](#) *Claviceps inconspicua* partial tef1a gene for translation elongation factor 1 alpha, strain CCC662

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 325C..... 346
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 592T..C 573

>[JQ425676.1](#) *Isaria* sp. BCC26302 translation elongation factor 1 alpha (tef) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 295T..... 316
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 562C..G 543

>[OP832394.1](#) *Trichoderma lentinulae* isolate JZBQT0Z2 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 813C..... 834
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1080C..G 1061

>[PV617230.1](#) *Trichoderma* sp. strain CFCC 80618 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 709C..... 730
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 976C..G 957

>[PV567539.1](#) *Lecanicillium aphanocladii* isolate JFRL 03-5292 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 298T..... 319

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 565C..G 546

>[ON101292.1](#) *Aciculosporium take* isolate SICAUCC 22-0014 translation elongation factor 1-alpha (tefA) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 300C..... 321

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 567T..A 548

>[OQ471099.1](#) *Paragliomastix Chiangraiensis* isolate CBS277.80B translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 241C..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 508C..G 489

>[KF747112.1](#) *Tolypocladium tropicale* strain MX337 translation elongation factor 1 alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 291C..... 312

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 558C..G 539

>[PQ202125.1](#) *Ophiocordyceps* sp. isolate 77Polchen_OP translation elongation factor 1-alpha (EF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 292C..... 313

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 559C..G 540

>[MT430125.1](#) *Claviceps purpurea* isolate LM475 translation elongation factor 1-alpha (TEF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 208C..... 229

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 475T..C 456

>[MZ552298.1](#) *Trichoderma hirsutum* strain T11#9-1 translation elongation factor 1 alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 302C..... 323

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 569C..G 550

>[KC998998.1](#) *Emericellopsis pallida* strain CBS 490.71 translation elongation factor 1-alpha (TEF1a) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 296T..... 317

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 563C..G 544

>[MN605883.1](#) *Trichoderma zelibre* strain CGMCC 3.19695 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 635T..... 656

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 902C..G 883

>[MT430159.1](#) *Claviceps purpurea* isolate LM334 translation elongation factor 1-alpha (TEF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 208C..... 229

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 475T..C 456

>[MN065771.1](#) *Albacillium hingganense* strain SGSF339 translation elongation factor 1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 240C..... 261

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 507C..C 488

>[KY646200.1](#) *Moelleriella thanathonensis* voucher MFLU:16-2922 translation elongation factor 1-

alpha (EF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 296T..... 317

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 563C..G 544

>[GU734752.1](#) *Cordyceps spегazzinii* strain ARSEF 7850 translation elongation factor 1 alpha (TEF) gene, partial sequence

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 809T..... 830

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1076C..G 1057

>[OP797918.1](#) *Aschersonia mackerrasiae* strain BRIP 62329a translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 285C..... 306

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 552T..G 533

>[FJ545737.1](#) *Beauveria bassiana* isolate EABb 04/03-Su translation elongation factor 1 alpha (tef1a) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 355T..... 376

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 622C..G 603

>[PV273492.1](#) *Ovicillium oosporum* strain CBS 112091 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 241C..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 508C..G 489

>[JN819015.1](#) *Ophiocordyceps unilateralis* isolate Co22 translation elongation factor 1 alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 328C..... 349

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 595C..G 576

>[MW748225.1](#) *Trichoderma atroviride* isolate T35 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 798C..... 819

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1065C..A 1046

>[LC529966.1](#) *Metapochonia suchlasporia* NBRC:32276 TEF1a gene for translation elongation factor 1 alpha, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 298T..... 319

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 565C..A 546

>[OQ944494.1](#) *Sesquicillium buxi* culture CBS:202.69 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 241T..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 508C..A 489

>[GU979952.1](#) *Cordyceps* sp. strain MY626 translation elongation factor 1 alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 295C..... 316

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 562C..C 543

>[MT430051.1](#) *Claviceps purpurea* isolate LM1018 translation elongation factor 1-alpha (TEF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 208C..... 229

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 475T..C 456

>[OQ927339.1](#) *Beauveria bassiana* strain NRID13 translation elongation factor-1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 247T..... 268

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 514C..G 495

>[KP455739.1](#) *Metarhizium pemphigi* strain RCEF1700 translation elongation factor 1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 290C..... 311

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 557C..C 538

>[MT249288.1](#) *Trichoderma aggressivum* f. *aggressivum* isolate 393 translation elongation factor 1-alpha (tef1) gene, exons 5 and 6 and partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 797C..... 818

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1064C..G 1045

>[OP832401.1](#) *Trichoderma paraviridescens* isolate JZBQT11Z1 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 844C..... 865

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1111C..A 1092

>[MH647788.1](#) *Trichoderma atroviride* isolate HZA13 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 320C..... 341

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 587C..A 568

>[ON649902.1](#) *Trichoderma* sp. WTQ-2022b isolate JZBQT5Z1 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 817C..... 838

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1084C..G 1065

>[OM482389.1](#) *Purpureocillium atypicola* strain RCEF1074 translation elongation factor 1-alpha (tef)

gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 295C..... 316

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 562C..A 543

>[MH986280.1](#) *Metarhizium brunneum* isolate AGF66 translation elongation factor 1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 233T..... 254

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 500T..C 481

>[MT430025.1](#) *Claviceps purpurea* isolate LM27 translation elongation factor 1-alpha (TEF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 208C..... 229

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 475T..C 456

>[XM_024912186.1](#) *Trichoderma harzianum* CBS 226.95 hypothetical protein (M431DRAFT_12328), mRNA

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 816C..... 837

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1083C..G 1064

>[MK955950.1](#) *Arachnidicola araneicola* isolate GY29011 translation elongation factor 1 alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 302T..... 323

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 569C..G 550

>[PQ202098.1](#) *Ophiocordyceps* sp. isolate 85Cryatra_OP translation elongation factor 1-alpha (EF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 292C..... 313

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 559C..G 540

>[PQ319908.1](#) *Emericellopsis atlantica* isolate SFC20240607-M005 translation elongation factor 1 (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 306T..... 327

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 573C..G 554

>[PQ659026.1](#) *Metarhizium pemphigi* strain NR-OM23 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 235C..... 256

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 502C..C 483

>[LC496907.1](#) *Simplicillium cylindrosporum* JCM 18170 TEF1a gene for translation elongation factor 1 alpha, partial sequence

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 208A..... 229

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 475C..G 456

>[MW267312.1](#) *Trichoderma harzianum* isolate TRA1-25 translation elongation factor 1-alpha (TEF-1 alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 789C..... 810

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1056C..G 1037

>[OQ470931.1](#) *Acremonium polychromum* isolate CBS181.27 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 241T..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 508C..G 489

>[LC769728.1](#) *Beauveria scarabaeidicola* BCRC:FU30836 tef gene for translation elongation factor 1-alpha, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 331T..... 352

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 598C..G 579

>[OR352911.1](#) *Akanthomyces lecanii* strain IRAN-3686C translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 305T..... 326

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 572C..G 553

>[OQ470962.1](#) *Hapsidospora globosa* isolate CBS514.70 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 241C..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 508C..G 489

>[MK504358.1](#) *Beauveria pseudobassiana* strain NREP099 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 245T..... 266

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 512C..G 493

>[KC998963.1](#) *Acremonium* sp. A104 translation elongation factor 1-alpha (TEF1a) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 290T..... 311

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 557C..A 538

>[MT430010.1](#) *Claviceps purpurea* isolate LM59 translation elongation factor 1-alpha (TEF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 208C..... 229

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 475T..C 456

>[GQ250026.1](#) *Akanthomyces arachnophilus* strain BCC17654 translation elongation factor 1 alpha

(tef) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 198T..... 219
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 465C..G 446

>OL440146.1 *Thyronectria abieticola* strain THYA translation elongation factor 1-alpha gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 827T..... 848
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1094C..G 1075

>MT428409.1 *Cordyceps cicadae* voucher JGS150805-3 clone Isa3 translation elongation factor 1 alpha (TEF) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 288T..... 309
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 555C..G 536

>OQ944501.1 *Sesquicillium buxi* strain JW199009 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 241T..... 262
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 508C..A 489

>OQ471135.1 *Protocreopsis rutila* isolate CBS394.70 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 241T..... 262
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 508C..G 489

>KY769931.1 *Digitiseta parvidigitata* strain MUCL 48180 translation elongation factor 1-alpha gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 302T..... 323
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 569C..A 550

>[KP008923.1](#) *Trichoderma gamsii* strain 37.24.06.1 translation elongation factor 1-alpha (tef1) gene, exons 5 and 6 and partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 756C..... 777

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1023C..A 1004

>[PQ630185.1](#) *Ophiocordyceps* sp. NS-2024a isolate GZUHB01H translation elongation factor 1-alpha (EF1alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 306T..... 327

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 573T..G 554

>[KJ500423.1](#) *Beauveria bassiana* strain ARSEF 5987 elongation factor-1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 955T..... 976

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1222C..G 1203

>[KJ665478.1](#) *Trichoderma europaeum* strain S114 voucher WU:33302 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 863T..... 884

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1130C..G 1111

>[OR548073.1](#) *Trichoderma guizhouense* isolate 526 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 819C..... 840

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1086C..G 1067

>[LT220751.1](#) *Beauveria bassiana* partial tef1a gene for translation elongation factor 1 alpha, strain GMSL06

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 294C..... 315

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 561C..G 542

>[MN785130.1](#) *Ophiocordyceps kimflemingiae* isolate SJS4_Ophio translation elongation factor 1-alpha gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 241C..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 508C..G 489

>[OQ858799.1](#) *Trichoderma aff. hortense* AD46-S3-3 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 793C..... 814

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1060C..G 1041

>[MN944388.1](#) *Simplicillium* sp. strain BCC88410 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 280A..... 301

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 547C..G 528

>[MT081437.1](#) *Trichoderma longibrachiatum* isolate Tloum3 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 847C..... 868

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1114C..A 1095

>[PP848437.1](#) *Trichoderma* sp. LT-2024a strain ZHKUCC 24-0781 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 795C..... 816

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1062C..G 1043

>[PV166499.1](#) *Gibellula penicillioides* strain RCEF7684 translation elongation factor 1-alpha (TEF1)

gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 258A..... 279

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 525C..A 506

>[MW267311.1](#) *Trichoderma harzianum* isolate TRA1-21 translation elongation factor 1-alpha (TEF-1 alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 789C..... 810

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1056C..G 1037

>[KU041509.1](#) *Sepedonium chalcipori* strain KSH602 translation elongation factor 1 alpha gene, exon 6 and partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 203C..... 224

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 470C..G 451

>[MT428411.1](#) *Cordyceps cicadae* voucher JGS150805-3 clone Cor3 translation elongation factor 1 alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 288T..... 309

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 555C..G 536

>[MZ422409.1](#) *Metarhizium lepidiotae* strain M1-01 translation elongation factor 1 alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 285T..... 306

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 552T..C 533

>[MF362082.1](#) *Cordyceps takaomontana* voucher ANDES-F 1255 elongation factor-1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 206T..... 227

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 473C..A 454

>[ON649899.1](#) *Trichoderma* sp. WTQ-2022f isolate JZBQT4Z1 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 816C..... 837

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1083C..G 1064

>[MN401491.1](#) *Beauveria bassiana* strain BCC34350 translation elongation factor 1 alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 313T..... 334

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 580C..G 561

>[KM091835.1](#) *Beauveria bassiana* strain GKKV 01_15 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 302T..... 323

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 569C..G 550

>[PV273676.1](#) *Ijuhya tetraspora* strain CBS 142016 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 241T..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 508C..G 489

>[KC633271.1](#) Uncultured *Lecanicillium* clone JCh021 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 147T..... 168

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 414C..G 395

>[ON951920.1](#) *Trichoderma* sp. culture NFCCI:5214 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 805C..... 826

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1072C..A 1053

>[OQ186391.1](#) *Ophiocordyceps sporangifera* voucher MFLU:22-0270 translation elongation factor 1-alpha (tef1a) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 274C..... 295

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 541C..G 522

>[KU933431.1](#) *Isaria sinclairii* strain ATCC 24400 translation elongation factor 1-alpha (EF1a) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 304C..... 325

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 571C..G 552

>[MK040102.1](#) *Beauveria bassiana* isolate B1 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 272T..... 293

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 539C..G 520

>[JQ425682.1](#) *Isaria* sp. BCC47987 translation elongation factor 1 alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 325T..... 346

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 592C..G 573

>[JN819019.1](#) *Ophiocordyceps unilateralis* isolate Co16 translation elongation factor 1 alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 338C..... 359

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 605C..G 586

>[KJ665532.1](#) *Trichoderma koningii* strain S28 translation elongation factor 1-alpha (tef1) gene,

partial cds

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product length = 268
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template       810    .....C.....  831

Reverse primer  1      GCTGCTCGTGGTGCATNTCN  20
Template       1077   .....C..G  1058

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>[KX713682.1](#) *Ophiocordyceps ootakii* strain J14 translation elongation factor 1 alpha (TEF) gene, partial cds

```

product length = 268
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template       319    .....C.....  340

Reverse primer  1      GCTGCTCGTGGTGCATNTCN  20
Template       586    .....C..G  567

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>[MT430179.1](#) *Claviceps purpurea* isolate Clav38 translation elongation factor 1-alpha (TEF1-alpha) gene, partial cds

```

product length = 268
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template       208    .....C.....  229

Reverse primer  1      GCTGCTCGTGGTGCATNTCN  20
Template       475    .....T..C  456

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>[KX344431.1](#) *Trichoderma verticillatum* strain HMAS 248740 translation elongation factor 1-alpha (tef1) gene, partial cds

```

product length = 268
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template       826    .....C.....  847

Reverse primer  1      GCTGCTCGTGGTGCATNTCN  20
Template       1093   .....C..G  1074

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>[EU420127.1](#) *Clavicipitaceae* sp. CV-2008a translation elongation factor 1 alpha gene, partial cds

```

product length = 271
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template       834    .....C.....  855

Reverse primer  1      GCTGCTCGTGGTGCATNTCN  20
Template       1104   .....C..G  1085

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>[KC285688.1](#) *Trichoderma paraviridescens* strain S68 translation elongation factor 1-alpha (tef1) gene, partial cds

```

product length = 268
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template       837    .....C.....  858

Reverse primer  1      GCTGCTCGTGGTGCATNTCN  20

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Template 1104C..A 1085

>[FJ860707.1](#) *Trichoderma subeffusum* strain CBS 120929 translation elongation factor 1 alpha (tef1) gene, partial sequence

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 845C..... 866

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 1112C..G 1093

>[PX071844.1](#) *Emericellopsis fuci* strain SFC20250301-M027 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 275C..... 296

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 542C..G 523

>[LC370832.1](#) *Ophiocordyceps* sp. gene for translation elongation factor 1-alpha, partial cds, strain: GniTKB04_OP

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 235C..... 256

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 502C..G 483

>[OR854253.1](#) *Moelleriella* sp. ZY-2023b voucher YHH 2308026 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 284T..... 305

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 551T..G 532

>[MT430187.1](#) *Claviceps purpurea* isolate Clav23 translation elongation factor 1-alpha (TEF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 208C..... 229

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 475T..C 456

>[PV617182.1](#) *Trichoderma* sp. strain CFCC 51939 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 710C..... 731

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 977C..G 958

>[OR066639.1](#) *Ophiocordyceps* sp. isolate TZEFB01 translation elongation factor 1-alpha (EF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 235C..... 256

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 502C..G 483

>[MF362057.1](#) *Cordyceps takaomontana* voucher ANDES-F 1259 elongation factor-1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 206T..... 227

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 473C..A 454

>[KP689555.1](#) *Epichloe festucae* strain FI1 translation elongation factor 1-alpha (tefA) gene, complete cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 1133T..... 1154

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 1400T..A 1381

>[OQ440736.1](#) *Ophiocordyceps liangshanensis* voucher KUMCC 3091 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 285C..... 306

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 552C..G 533

>[MZ703301.1](#) *Beauveria bassiana* isolate FRh2 elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 230T..... 251

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 497C..G 478

>[OQ944527.1](#) *Sesquicillium neerlandicum* culture CBS:148212 translation elongation factor 1-alpha

(tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 241T..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 508C..A 489

>LC163540.1 *Beauveria asiatica* EF1-a gene for translation elongation factor 1 alpha, partial cds, isolate: F402

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 326T..... 347

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 593C..G 574

>OQ944586.1 *Clonostachys divergens* culture CBS:532.69 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 241T..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 508C..A 489

>OQ944723.1 *Clonostachys rogersoniana* culture CBS:920.97 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 230A..... 251

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 497C..A 478

>FJ860678.1 *Hypocrea protopulvinata* strain C.P.K. 2476 translation elongation factor 1 alpha (tef1) gene, partial sequence

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 832C..... 853

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1099C..A 1080

>OR667283.1 *Trichoderma harzianum* culture SZMC:1643 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 815C..... 836

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 1082C..G 1063

>[MT429985.1](#) *Claviceps purpurea* isolate LM211 translation elongation factor 1-alpha (TEF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 208C..... 229

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 475T..C 456

>[PX071859.1](#) *Purpureocillium* sp. strain SFC20250301-M063 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 297C..... 318

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 564C..A 545

>[KU051703.1](#) *Trichoderma auranteffusum* strain 9143 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 885T..... 906

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1152C..G 1133

>[MT430027.1](#) *Claviceps purpurea* isolate LM24 translation elongation factor 1-alpha (TEF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 208C..... 229

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 475T..C 456

>[MH092896.1](#) *Ophiocordyceps houaynhangensis* strain MY10729 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 325C..... 346

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 592C..G 573

>[ON398009.1](#) *Beauveria bassiana* isolate Bb3 translation elongation factor gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 346T..... 367

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 613C..G 594

>[OQ927332.1](#) *Beauveria bassiana* strain NRID06 translation elongation factor-1 alpha gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 268T..... 289

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 535C..G 516

>[PV273610.1](#) *Clavatomyces korffii* strain CBS 138710 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 241C..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 508C..G 489

>[OQ470844.1](#) *Emericellopsis fimetaria* isolate CBS621.85 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 241C..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 508C..G 489

>[ON101291.1](#) *Aciculosporium take* isolate SICAUCC 22-0013 translation elongation factor 1-alpha (tefA) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 293C..... 314

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 560T..A 541

>[AY489610.1](#) *Balansia henningsiana* strain AEG96-27a translation elongation factor 1 alpha gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 328C..... 349

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 595T..A 576

>[MW267369.1](#) *Trichoderma harzianum* isolate TRA2-27 translation elongation factor 1-alpha (TEF-

1 alpha) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 789C..... 810
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1056C..G 1037

>PQ650475.1 Clonostachys sp. JM-2024c strain HSC398 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 303T..... 324
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 570C..A 551

>HM534879.1 Thyronectria sinopica strain CBS 127386 translation elongation factor 1 alpha (tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 800A..... 821
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1067C..A 1048

>OP693448.1 Beauveria bassiana strain LPSc1489 translation elongation factor 1-alpha gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 292C..... 313
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 559C..G 540

>MH057732.1 Hirsutella eleutheratorum culture ARSEF:13375 translation elongation factor 1-alpha gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 298C..... 319
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 565C..G 546

>KY688027.1 Trichoderma concentricum voucher HMAS:248833 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 817C..... 838
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 1084C..A 1065

>[ON326404.1](#) *Hevansia* sp. TW-2022a voucher NL20210822-03 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 295T..... 316

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 562C..G 543

>[PV273522.1](#) *Emericellopsis proliferata* strain CBS 228.59 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 241T..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 508C..G 489

>[OQ470879.1](#) *Emericellopsis moniliformis* isolate CBS864.73 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 241C..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 508C..G 489

>[MZ955880.1](#) *Papiliomyces* sp. YZ-2021a voucher YC20061403 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 220T..... 241

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 487C..C 468

>[MK040106.1](#) *Beauveria bassiana* isolate B12 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 272T..... 293

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 539C..G 520

>[KJ665676.1](#) *Trichoderma polysporum* strain S176 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 859T..... 880

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1126C..G 1107

>[PV290064.1](#) *Trichoderma harzianum* isolate Endo3 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 638C..... 659

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 905C..G 886

>[EF193181.1](#) *Beauveria bassiana* isolate KVL 03_114 translation elongation factor 1 alpha gene, partial sequence

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 1057T..... 1078

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1324C..G 1305

>[PV617202.1](#) *Trichoderma* sp. strain CFCC 53271 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 714C..... 735

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 981C..A 962

>[MH244848.1](#) *Purpureocillium lilacinum* strain AMEF095 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 296C..... 317

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 563C..A 544

>[MT249298.1](#) *Trichoderma aggressivum* f. *aggressivum* isolate Boz1 translation elongation factor 1-alpha (tef1) gene, exons 5 and 6 and partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 797C..... 818

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1064C..G 1045

>[PP600951.1](#) *Trichoderma paraviridescens* isolate T20C14 translation elongation factor 1-alpha

(tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 767C..... 788

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1034C..A 1015

>MK491909.1 Ophiocordyceps sp. voucher Gh06 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 290C..... 311

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 557T..G 538

>DQ070001.1 Aschersonia aleyrodis strain PC578 elongation factor 1-alpha-like gene, partial sequence

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 193T..... 214

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 460C..G 441

>GU797120.1 Ophiocordyceps brunneipunctata strain NHJ1491 translation elongation factor 1 alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 295C..... 316

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 562C..C 543

>OQ470926.1 Gliomastix musae isolate CBS617.94 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 241C..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 508C..G 489

>LC684117.1 Claviceps purpurea TNS:F-96394 TEF gene for translation elongation factor 1-alpha, partial sequence

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 326C..... 347

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 593T..C 574

>[MW267434.1](#) *Trichoderma harzianum* isolate TB2-13 translation elongation factor 1-alpha (TEF-1 alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 789C..... 810

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1056C..G 1037

>[OQ471203.1](#) *Stephanonectria keithii* isolate CBS943.72 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 241C..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 508C..A 489

>[OL322031.1](#) *Ascopolyporus galloides* strain BCC48704 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 286C..... 307

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 553C..G 534

>[MN401475.1](#) *Beauveria asiatica* strain BCC76509 translation elongation factor 1 alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 235T..... 256

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 502C..G 483

>[FJ860685.1](#) *Trichoderma rhododendri* strain CBS 119288 translation elongation factor 1 alpha (tef1) gene, partial sequence

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 880C..... 901

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1147C..G 1128

>[AY986926.1](#) *Aschersonia andropogonis* strain P.C. 535 translation elongation factor 1-alpha (EF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 289A..... 310

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 556C..G 537

>[KC339704.1](#) *Beauveria hoplocheli* strain Bt121 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 1071T..... 1092

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 1338C..G 1319

>[KT878369.1](#) *Acremonium sclerotigenum* strain 06238 translation elongation factor 1-alpha (TEF1a) gene, partial cds

product length = 271

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 286C..... 307

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 556C..G 537

>[MZ703295.1](#) *Beauveria caledonica* isolate F532 elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 230T..... 251

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 497C..G 478

>[MF362090.1](#) *Cordyceps takaomontana* voucher ANDES-F 1312 elongation factor-1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 179T..... 200

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 446C..A 427

>[KM651993.1](#) *Hirsutella fusiformis* voucher ARSEF 5474 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 298C..... 319

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 565C..G 546

>[KT149300.1](#) *Trichoderma crassum* strain 116 translation elongation factor 1-alpha (tef1) gene,

partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 631C..... 652

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 898C..G 879

>[LC598982.1](#) *Claviceps* sp. TNS:F60495 Ef1-alpha gene for translation elongation factor 1-alpha, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 326C..... 347

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 593T..C 574

>[OM482378.1](#) *Cordyceps javanica* strain RCEF2413 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 293T..... 314

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 560C..G 541

>[PV273475.1](#) *Stilbocrea* sp. strain CBS 658.83 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 241C..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 508C..C 489

>[KJ665688.1](#) *Trichoderma priscilae* strain S118 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 818C..... 839

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1085C..G 1066

>[MN094771.1](#) *Beauveria peruvienensis* isolate UTRF35 translation elongation factor 1 alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 720T..... 741

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 987C..G 968

>[FJ860643.1](#) *Trichoderma longipile* strain CBS 120953 translation elongation factor 1 alpha (tef1) gene, partial sequence

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 846C..... 867

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1113C..G 1094

>[OP820494.1](#) *Simplicillium lanosoniveum* voucher CHE-CNRCB-544 translation elongation factor 1 alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 294A..... 315

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 561C..G 542

>[EF468773.1](#) *Cordyceps* sp. OSC 110996 translation elongation factor 1 alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 328C..... 349

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 595T..C 576

>[KM091837.1](#) *Beauveria bassiana* strain GKKV 01_31 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 300T..... 321

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 567C..G 548

>[MF416483.1](#) *Cordyceps* sp. voucher MCA 1727 translation elongation factor 1 alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 292T..... 313

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 559C..G 540

>[MH203491.1](#) *Beauveria bassiana* strain CHE-CNRCB 430 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 315T..... 336

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 582C..G 563

>[MN401457.1](#) *Beauveria asiatica* strain BCC36985 translation elongation factor 1 alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 319T..... 340

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 586C..G 567

>[EU248852.1](#) *Metarhizium anisopliae* culture-collection ARSEF:7450 translation elongation factor 1 alpha (TEF) gene, partial sequence

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 952T..... 973

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1219T..C 1200

>[OQ944482.1](#) *Nectriopsis didymii* culture CBS:326.79 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 241A..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 508C..G 489

>[MK336006.1](#) *Amphichorda cavernicola* strain LC12674 translation elongation factor 1-alpha (TEF1a) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 293C..... 314

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 560C..G 541

>[MH062193.1](#) *Lecanicillium* sp. culture CEP:419 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 283T..... 304

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 550C..G 531

>[MT929245.1](#) *Trichoderma afroharzianum* strain AG500 translation elongation factor 1-alpha

(EF1a) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 297C..... 318
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 564C..G 545

>PQ166335.1 Metarhizium sp. strain URM 6133 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 285T..... 306
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 552T..C 533

>OR917447.1 Amphichorda felina strain CBS 250.34 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 296C..... 317
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 563C..G 544

>MH244885.1 Beauveria bassiana strain AMEF006 translation elongation factor 1-alpha gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 248T..... 269
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 515C..G 496

>ON897693.1 Trichoderma sp. SL-2022a voucher GMB0011 translation elongation factor 1-alpha gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 751C..... 772
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1018C..A 999

>OQ470845.1 Emericellopsis fimetaria isolate CBS628.85 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 241C..... 262
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 508C..G 489

>[PV755814.1](#) *Trichoderma* sp. strain CFCC 53270A translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 831T..... 852

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1098C..G 1079

>[OQ471091.1](#) *Paracylindrocarpon aloicola* isolate CBS335.77 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 241C..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 508C..A 489

>[PQ283815.1](#) *Cordyceps cateniannulata* isolate CcQZ-03 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 299T..... 320

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 566C..G 547

>[OQ944614.1](#) *Clonostachys farinosa* culture CBS:202.93 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 241T..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 508C..G 489

>[KJ665771.1](#) *Trichoderma viride* strain S552 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 848C..... 869

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1115C..G 1096

>[KX344435.1](#) *Trichoderma hebeiense* strain HMAS 248744 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 840C..... 861

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1107C..G 1088

>[OR066658.1](#) *Ophiocordyceps* sp. isolate FSEFB01 translation elongation factor 1-alpha (EF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 235C..... 256

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 502C..G 483

>[MT430184.1](#) *Claviceps purpurea* isolate Clav27 translation elongation factor 1-alpha (TEF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 208C..... 229

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 475T..C 456

>[OR832089.1](#) *Hypocrella* sp. ZY-2023a voucher YHH 2303015 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 293C..... 314

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 560C..G 541

>[KJ398799.1](#) *Metarhizium minus* voucher ARSEF 1099 translation elongation factor 1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 1056C..... 1077

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1323T..C 1304

>[MG595724.1](#) *Trichoderma asperellum* isolate B1581 translation elongation factor 1-alpha (tef1alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 731C..... 752

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 998C..G 979

>[XM_073706325.1](#) *Trichoderma ghanense* translation elongation factor EF-1 alpha

(CCMA1212_009234), partial mRNA

```

product length = 268
Forward primer  1    TGGTACAAGGGNTGGGAGAAGG  22
Template        595  .....C.....  616

Reverse primer  1    GCTGCTCGTGGTGCATNTCN  20
Template        862  .....C..G  843

```

>[PP848441.1](#) *Trichoderma* sp. LT-2024e strain ZHKUCC 24-0785 translation elongation factor 1-alpha (TEF1) gene, partial cds

```

product length = 268
Forward primer  1    TGGTACAAGGGNTGGGAGAAGG  22
Template        821  .....C.....  842

Reverse primer  1    GCTGCTCGTGGTGCATNTCN  20
Template       1088  .....C..G  1069

```

>[OQ858756.1](#) *Trichoderma virens* isolate BN47-S5-5 translation elongation factor 1-alpha (tef1) gene, partial cds

```

product length = 268
Forward primer  1    TGGTACAAGGGNTGGGAGAAGG  22
Template        796  .....C.....  817

Reverse primer  1    GCTGCTCGTGGTGCATNTCN  20
Template       1063  .....C..G  1044

```

>[MT430172.1](#) *Claviceps purpurea* isolate Clav45 translation elongation factor 1-alpha (TEF1-alpha) gene, partial cds

```

product length = 268
Forward primer  1    TGGTACAAGGGNTGGGAGAAGG  22
Template        208  .....C.....  229

Reverse primer  1    GCTGCTCGTGGTGCATNTCN  20
Template        475  .....T..C  456

```

>[GU797126.1](#) *Ophiocordyceps dipterigena* strain MY621 translation elongation factor 1 alpha (tef) gene, partial cds

```

product length = 268
Forward primer  1    TGGTACAAGGGNTGGGAGAAGG  22
Template        274  .....C.....  295

Reverse primer  1    GCTGCTCGTGGTGCATNTCN  20
Template        541  .....T..A  522

```

>[DQ307524.2](#) *Trichoderma trixiae* strain GJS 92-11 elongation factor 1 alpha (tef1) gene, partial sequence

```

product length = 268
Forward primer  1    TGGTACAAGGGNTGGGAGAAGG  22
Template        844  .....C.....  865

Reverse primer  1    GCTGCTCGTGGTGCATNTCN  20

```

Template 1111C..A 1092

>[OR115173.1](#) *Metapochonia suchlasporia* strain YFCC 807 translation elongation factor 1 alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 328T..... 349

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 595C..A 576

>[OR553412.1](#) *Beauveria bassiana* voucher CHE-CNRCB 1193 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 302T..... 323

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 569C..G 550

>[LT216545.1](#) *Claviceps* sp. CCC 1240 partial tef1a gene for translation elongation factor 1 alpha, strain CCC 1240

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 325T..... 346

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 592T..C 573

>[HQ722026.1](#) *Moelleriella pumatensis* isolate BCC41004 translation elongation factor 1-alpha (EF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 283T..... 304

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 550T..G 531

>[KJ665735.1](#) *Trichoderma* sp. S467 culture CBS:135582 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 814C..... 835

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 1081C..G 1062

>[LT216521.1](#) *Claviceps maximensis* partial tef1a gene for translation elongation factor 1 alpha, strain CCC 816

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 325C..... 346

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 592T..C 573

>[OQ471999.1](#) *Metarhizium brunneum* strain BUM 1041 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 299T..... 320

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 566T..C 547

>[MF614643.1](#) *Ophiocordyceps brunneaperitheciata* strain BCC64201 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 283G..... 304

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 550C..G 531

>[MT386535.1](#) *Tolypocladium* sp. strain YFCC 3009 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 284C..... 305

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 551C..G 532

>[EF468764.1](#) *Ophiocordyceps rhizoidea* strain NHJ 12522 translation elongation factor 1 alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 328C..... 349

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 595C..G 576

>[OR115219.1](#) *Keithomyces* sp. QZW-2023b strain YFCC 850.01 translation elongation factor 1 alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 328C..... 349

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 595T..C 576

>[PV273619.1](#) *Geonectria quercus* strain CBS 137308 translation elongation factor 1-alpha (TEF1)

gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 241A..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 508C..G 489

>[MT184202.1](#) *Claviceps purpurea* strain YFCC 8511 translation elongation factor 1-alpha (TEF)
 gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 325C..... 346

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 592T..C 573

>[PQ650477.1](#) *Clonostachys linzhiensis* strain YHD691 translation elongation factor 1-alpha (tef1)
 gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 139T..... 160

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 406C..A 387

>[MW267397.1](#) *Trichoderma harzianum* isolate TA2-10 translation elongation factor 1-alpha (TEF-1
 alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 806C..... 827

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1073C..G 1054

>[MG383493.1](#) *Trichoderma* sp. 3 WYZ-2018 strain HMAS 275665 translation elongation factor 1-
 alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 814C..... 835

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1081C..G 1062

>[OQ944583.1](#) *Clonostachys divergens* culture CBS:102426 translation elongation factor 1-alpha
 (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 241T..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 508C..A 489

>[JN715658.1](#) *Trichoderma hispanicum* strain S423 translation elongation factor 1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 835C..... 856

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1102C..G 1083

>[EU248866.1](#) *Metarhizium majus* culture-collection ARSEF:1015 translation elongation factor 1 alpha (TEF) gene, partial sequence

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 1032T..... 1053

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1299T..C 1280

>[PV755811.1](#) *Trichoderma* sp. strain CFCC 53257A translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 824C..... 845

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1091C..A 1072

>[MW344961.1](#) *Metarhizium anisopliae* isolate NCHU-95 translation elongation factor 1-alpha (TEF-1alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 312T..... 333

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 579T..C 560

>[OR965920.1](#) *Metarhizium* sp. isolate MGV7 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 351T..... 372

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 618T..C 599

>[JX192799.1](#) *Hevansia novoguineensis* strain EPF046 translation elongation factor 1 alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 295T..... 316

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 562C..G 543

>[FJ860698.1](#) *Hypocrea sinuosa* strain C.P.K. 2008 translation elongation factor 1 alpha (tef1) gene, partial sequence

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 852C..... 873

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1119C..G 1100

>[JX083696.1](#) *Claviceps microcephala* isolate CCC480 translation elongation factor 1-alpha (EF1-alpha) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 325C..... 346

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 592T..C 573

>[MH143859.1](#) *Metarhizium viridulum* isolate BUM_721 translation elongation factor 1-alpha gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 297T..... 318

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 564T..C 545

>[ON934340.1](#) *Trichoderma afroharzianum* isolate a91 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 803C..... 824

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1070C..G 1051

>[OR672130.1](#) *Cordyceps javanica* strain SS-L3-14F translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 304T..... 325

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 571C..G 552

>[OQ743421.1](#) *Cordyceps* sp. voucher KUMCC 9351 translation elongation factor 1-alpha (tef) gene,

partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 285C..... 306

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 552C..T 533

>[ON125018.1](#) *Niveomyces formicidarum* isolate BCC79346 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 283C..... 304

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 550C..G 531

>[KT225598.1](#) *Isaria javanica* strain CHE-CNRCB 303 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 340T..... 361

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 607C..G 588

>[KJ665660.1](#) *Trichoderma petersenii* strain S500 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 817C..... 838

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1084C..G 1065

>[PV273578.1](#) *Stephanonectria chromolaenae* strain CBS 102562 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 241C..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 508C..A 489

>[MK850835.1](#) *Trichoderma atroviride* isolate HZA13 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 822C..... 843

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 1089C..A 1070

>[OQ470916.1](#) Gliomastix murorum isolate CBS157.72 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 241C..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 508C..G 489

>[KF747098.1](#) Tolypocladium RG-2013a strain MS271 translation elongation factor 1 alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 250C..... 271

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 517C..C 498

>[JN998771.1](#) Isaria fumosorosea strain CNPSo162 translation elongation factor 1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 328C..... 349

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 595C..A 576

>[MW116359.1](#) Trichoderma citrinoviride strain 18CPHA003 translation elongation factor 1 alpha (TEF1a) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 873C..... 894

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1140C..G 1121

>[PV617190.1](#) Trichoderma sp. strain CFCC 53193 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 721C..... 742

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 988C..G 969

>[KJ665417.1](#) Trichoderma atroviride strain S264 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 821C..... 842

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1088C..A 1069

>[JQ867163.1](#) *Beauveria bassiana* strain RCEF4676 elongation factor-1 alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 305T..... 326

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 572C..G 553

>[KJ536063.1](#) *Beauveria varroae* strain EABv 09/02-Fil translation elongation factor 1-alpha (tef1a) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 321T..... 342

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 588C..G 569

>[KX632631.1](#) *Trichoderma alni* strain T16 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 795C..... 816

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1062C..A 1043

>[EU718260.1](#) *Aschersonia* sp. WYTM-1 translation elongation factor 1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 352T..... 373

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 619T..G 600

>[KJ665755.1](#) *Trichoderma tomentosum* strain S193 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 811C..... 832

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1078C..G 1059

>[MT430103.1](#) *Claviceps purpurea* isolate LM561 translation elongation factor 1-alpha (TEF1-alpha)

gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 208C..... 229

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 475T..C 456

>[PP740440.1](#) Lasionectriella rubioi isolate SNC201 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 298C..... 319

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 565C..C 546

>[OQ944881.1](#) Clonostachys zelandiae-novae culture CBS:102422 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 241T..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 508C..A 489

>[MH070615.1](#) Beauveria bassiana isolate BsoilAR4.1-2 elongation factor-1 alpha (EF1-a) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 269T..... 290

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 536C..G 517

>[OR188675.1](#) Beauveria pseudobassiana isolate NRXG17 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 244T..... 265

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 511C..G 492

>[KM283821.1](#) Lecanicillium muscarium strain CBS 143.62 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 289T..... 310

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 556C..G 537

>[KJ665476.1](#) *Trichoderma europaeum* strain Hypo 64 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 870T..... 891

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1137C..G 1118

>[KX428018.1](#) *Trichoderma* sp. TC162 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 593C..... 614

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 860C..G 841

>[OM471991.1](#) *Trichoderma asperellum* isolate Tbk translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 298C..... 319

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 565C..G 546

>[MZ360967.1](#) *Sepedonium ampullosporium* strain CONC-F 1736 elongation factor 1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 130C..... 151

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 397C..G 378

>[AY489616.1](#) *Cordyceps gunnii* translation elongation factor 1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 328C..... 349

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 595C..A 576

>[PV755848.1](#) *Trichoderma* sp. strain CFCC 81907A translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 796C..... 817

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1063C..G 1044

>[DQ070026.1](#) *Aschersonia andropogonis* strain PC451 elongation factor 1-alpha-like gene, partial sequence

product length = 267

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 185A..... 206

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 451T..G 432

>[MK015641.1](#) *Beauveria bassiana* voucher LPSc1216 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 293T..... 314

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 560C..G 541

>[MN094780.1](#) *Beauveria peruviana* isolate UTRP17 translation elongation factor 1 alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 720T..... 741

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 987C..G 968

>[KJ665770.1](#) *Trichoderma viride* strain S472 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 844C..... 865

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1111C..G 1092

>[KJ665359.1](#) *Trichoderma atroviride* strain Hypo 25 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 815C..... 836

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1082C..G 1063

>[KR264017.1](#) *Hirsutella rhossiliensis* strain ARSEF3746 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 232C..... 253

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 499C..A 480

>[MK632064.1](#) *Metarhizium phasmatodeae* strain BCC49272 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 289T..... 310

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 556T..C 537

>[KT343760.1](#) *Trichoderma priscilae* voucher HMAS 245002 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 818C..... 839

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1085C..G 1066

>[MH183185.1](#) *Trichoderma* sp. XD-2018h strain YMF1.00351 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 791T..... 812

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1058C..G 1039

>[MH244872.1](#) *Cordyceps fumosorosea* strain AMEF094 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 250C..... 271

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 517C..A 498

>[AY391966.1](#) *Trichoderma chlorosporum* strain G.J.S. 88-33 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 183C..... 204

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 450C..G 431

>[OQ281359.1](#) *Acremonium hennebertii* isolate SICAUCC 22-0112 translation elongation factor 1-

alpha (tef1) gene, partial cds

product length = 271
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 287C..... 308
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 557C..G 538

>[OQ470739.1](#) Acremonium alternatum isolate CBS407.66 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 241T..... 262
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 508C..A 489

>[KJ665567.1](#) Trichoderma mediterraneum strain S184 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 829T..... 850
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1096C..G 1077

>[OP832406.1](#) Trichoderma auriculariae isolate JZBQT10Z10 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 816C..... 837
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1083C..G 1064

>[MN401498.1](#) Beauveria bassiana strain BCC1848 translation elongation factor 1 alpha (tef) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 319T..... 340
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 586C..G 567

>[KT878363.1](#) Acremonium sclerotigenum strain 07603 translation elongation factor 1-alpha (TEF1a) gene, partial cds

product length = 271
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 292C..... 313
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 562C..G 543

>[MN094777.1](#) *Beauveria peruviansis* isolate UTRP6 translation elongation factor 1 alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 720T..... 741

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 987C..G 968

>[PV617213.1](#) *Trichoderma* sp. strain CFCC 54640 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 711C..... 732

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 978C..A 959

>[MZ711216.1](#) *Beauveria bassiana* strain RCEF6978 translation elongation factor-1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 309T..... 330

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 576C..G 557

>[MF095879.1](#) *Trichoderma longifialidicum* strain TC668 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 807C..... 828

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1074C..G 1055

>[MG993029.1](#) *Acremonium sclerotigenum* translation elongation factor 1-alpha (TEF1a) gene, partial cds

product length = 271

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 300C..... 321

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 570C..G 551

>[PP855551.1](#) *Matutiniella brasiliensis* strain CDA 3357 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 292C..... 313

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 559C..C 540

>[KX713674.1](#) *Ophiocordyceps camponoti-floridani* strain Flx2 translation elongation factor 1 alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 262C..... 283

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 529C..G 510

>[OQ944642.1](#) *Clonostachys hongkongensis* culture CBS:115291 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 241T..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 508C..A 489

>[OR855790.1](#) *Ophiocordyceps brunneipunctata* strain BCC43276 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 252C..... 273

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 519C..C 500

>[MT430136.1](#) *Claviceps purpurea* isolate LM462 translation elongation factor 1-alpha (TEF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 208C..... 229

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 475T..C 456

>[PQ758604.1](#) *Amphichorda kunmingensis* strain CBS 312.50 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 328C..... 349

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 595C..G 576

>[MK491907.1](#) *Ophiocordyceps* sp. voucher BDS32 translation elongation factor 1-alpha gene,

partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 284C..... 305
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 551T..G 532

>[MK504360.1](#) *Beauveria pseudobassiana* isolate AMEF003 translation elongation factor 1-alpha gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 244T..... 265
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 511C..G 492

>[OR854258.1](#) *Moelleriella* sp. ZY-2023e voucher YHH 2308030 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 296A..... 317
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 563T..G 544

>[MK632061.1](#) *Metarhizium gryllicicola* strain BCC22353 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 289T..... 310
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 556T..C 537

>[MH244852.1](#) *Cordyceps fumosorosea* strain AMEF071 translation elongation factor 1-alpha gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 252C..... 273
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 519C..A 500

>[AY489615.1](#) *Elaphocordyceps capitata* translation elongation factor 1 alpha gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 328C..... 349
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 595C..C 576

>[OQ472002.1](#) *Beauveria brongniartii* strain BUB 1035 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 296T..... 317

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 563C..G 544

>[KJ398796.1](#) *Metarhizium khaoyaiense* voucher BCC 12687 translation elongation factor 1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 1030C..... 1051

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1297C..C 1278

>[LN846858.1](#) *Claviceps pazoutovae* partial tef1a gene for translation elongation factor 1 alpha, culture collection CCC:1393

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 325C..... 346

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 592T..C 573

>[MW723084.1](#) *Beauveria brongniartii* strain RCEF3224 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 295T..... 316

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 562C..G 543

>[MH244893.1](#) *Beauveria bassiana* strain AMEF098 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 294T..... 315

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 561C..G 542

>[MK984570.1](#) *Tolypocladium reniformisporum* strain YFCC 1805002 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 328C..... 349

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 595C..G 576

>[KJ398817.1](#) *Cordyceps cylindrica* voucher ARSEF 7231 translation elongation factor 1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 1163C..... 1184

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 1430C..G 1411

>[MW723095.1](#) *Cordyceps cicadae* strain RCEF4954 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 295T..... 316

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 562C..G 543

>[LC163549.1](#) *Beauveria asiatica* EF1-a gene for translation elongation factor 1 alpha, partial cds, isolate: F2849

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 326T..... 347

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 593C..G 574

>[ON010785.1](#) *Trichoderma virens* strain LZ003 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 803C..... 824

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 1070C..G 1051

>[PV167693.1](#) *Ophiocordyceps jilinensis* voucher GMB 3071 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 328C..... 349

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 595C..G 576

>[KJ665618.1](#) *Trichoderma minutisporum* culture CBS:112255 strain G.J.S. 90-82 translation

elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 868T..... 889

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1135C..G 1116

>[AY883700.1](#) *Beauveria bassiana* strain ARSEF 6234 translation elongation factor 1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 980T..... 1001

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1247C..G 1228

>[KJ665629.1](#) *Trichoderma paraviridescens* strain S664 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 844C..... 865

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1111C..A 1092

>[MW292433.1](#) *Ophiocordyceps tessaratomidarum* isolate MY10827 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 277T..... 298

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 544T..G 525

>[FJ860699.1](#) *Hypocrea spinulosa* strain CBS 121280 translation elongation factor 1 alpha (tef1) gene, partial sequence

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 841C..... 862

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1108C..A 1089

>[KC285665.1](#) *Trichoderma viridarium* strain S95 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 836C..... 857

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 1103C..A 1084

>[PP392590.1](#) *Emericellopsis minima* strain FMR 19461 translation elongation factor 1-alpha (Tef1a) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 299C..... 320

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 566C..A 547

>[ON512612.1](#) *Tolypocladium* sp. isolate JMS372 translation elongation factor EF1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 244C..... 265

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 511C..C 492

>[PV153477.1](#) *Moelleriella* sp. voucher XXL202010001-1 translation elongation factor 1-alpha (tef1) gene, partial cds; mitochondrial

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 293T..... 314

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 560C..G 541

>[OR296650.1](#) *Metarhizium pingshaense* strain ARSEF 7422 translation elongation factor 1-alpha (EF1a) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 277T..... 298

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 544T..C 525

>[MN068368.1](#) *Claviceps ripicola* voucher DAOMC:252149 translation elongation factor 1-alpha (Ef1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 197G..... 218

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 464C..C 445

>[PQ787212.1](#) *Ophiocordyceps* sp. isolate GZUIFR-2010MC-1 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 301T..... 322

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 568C..G 549

>[PX071845.1](#) *Emericellopsis fuci* strain SFC20250301-M028 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 204C..... 225

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 471C..G 452

>[KJ665516.1](#) *Trichoderma helicolixii* culture CBS:135583 strain S515 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 806T..... 827

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1073C..G 1054

>[EF468781.1](#) *Torrubiella alba* strain CBS 726.73a translation elongation factor 1 alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 328T..... 349

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 595C..G 576

>[JX083705.1](#) *Claviceps humidiphila* isolate CCC503 translation elongation factor 1-alpha (EF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 325C..... 346

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 592T..C 573

>[FJ860625.1](#) *Trichoderma margaretense* strain C.P.K. 3127 translation elongation factor 1 alpha (tef1) gene, partial sequence

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 895C..... 916

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1162C..G 1143

>[KJ435086.1](#) *Conoideocrella luteorostrata* strain BCC 9617 translation elongation factor 1-alpha

(tef) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 283T..... 304
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 550C..A 531

>OM482394.1 Purpureocillium atypicola strain RCEF3667 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 295C..... 316
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 562C..A 543

>OR066644.1 Ophiocordyceps sp. isolate PINFB01 translation elongation factor 1-alpha (EF1-alpha) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 235C..... 256
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 502C..G 483

>MN094770.1 Beauveria peruvienensis isolate UTRF26 translation elongation factor 1 alpha (tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 720T..... 741
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 987C..G 968

>KJ665536.1 Trichoderma koningii strain S346 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 816C..... 837
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1083C..G 1064

>OQ116920.1 Ophiocordyceps duyunensis strain HKAS125843 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 273C..... 294
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 540C..G 521

>[MN401447.1](#) *Beauveria loeiensis* strain BCC66202 translation elongation factor 1 alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 319T..... 340

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 586C..G 567

>[PV670067.1](#) *Metarhizium anisopliae* isolate IIHRMa13 translation elongation factor 1-alpha (TEF1a) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 268T..... 289

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 535T..C 516

>[KJ634755.1](#) *Trichoderma koningiopsis* isolate 7745 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 840C..... 861

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 1107C..G 1088

>[OQ067550.1](#) *Beauveria bassiana* isolate BLM11KK20 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 320T..... 341

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 587C..G 568

>[KU729729.1](#) *Keithomyces neogunnii* isolate GZUHSB13050305 translation elongation factor 1-alpha (EF1a) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 293C..... 314

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 560T..C 541

>[ON494580.1](#) *Acremonium globosporum* strain GZUIFR22.036 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 271

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 302C..... 323

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 572C..G 553

>[KX855229.1](#) *Metapochonia variabilis* strain LC5717 translation elongation factor 1-alpha (TEF1a) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 299T..... 320

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 566C..A 547

>[KT224464.1](#) *Trichoderma* sp. 2 WTQ-2015 strain 8889 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 669C..... 690

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 936C..G 917

>[DQ522319.1](#) *Balansia pilulaeformis* strain AEG 94-2 translation elongation factor 1-alpha (EF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 328C..... 349

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 595T..A 576

>[OR804511.1](#) *Beauveria bassiana* isolate 1SEL22 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 323C..... 344

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 590C..G 571

>[PQ826969.1](#) *Memnoniella* sp. KH-2024a voucher GMB6206 translation elongation factor 1-alpha (tef1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 289C..... 310

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 556C..G 537

>[PV741072.1](#) *Metarhizium anisopliae* isolate IIHRMa5 translation elongation factor 1-alpha gene,

partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 253T..... 274

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 520T..C 501

>[MZ552303.1](#) *Trichoderma gamsii* strain T5#4-3 translation elongation factor 1 alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 302C..... 323

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 569C..A 550

>[PV832505.1](#) *Beauveria bassiana* isolate TP17 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 319T..... 340

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 586C..G 567

>[OP562772.1](#) *Trichoderma simmonsii* isolate HF1S03 elongation factor 1-alpha (tef-1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 842C..... 863

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1109C..G 1090

>[KJ665752.1](#) *Trichoderma thelephoricola* strain S572 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 812C..... 833

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1079C..G 1060

>[KR049188.1](#) *Metarhizium pemphigi* strain RCEF1990 translation elongation factor 1 alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 290C..... 311

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 557C..C 538

>[PV751034.1](#) *Beauveria bassiana* strain IHRBb6 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 253T..... 274

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 520C..G 501

>[KY587209.1](#) *Beauveria bassiana* strain CHE-CNRCB 167 translation elongation factor 1 alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 336T..... 357

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 603C..G 584

>[KJ665622.1](#) *Trichoderma neorufoides* strain S568 voucher WU:33408 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 857C..... 878

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1124C..G 1105

>[OQ621347.1](#) *Beauveria bassiana* isolate LPSc1417 translation elongation factor 1-alpha (EF1alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 283C..... 304

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 550C..G 531

>[OU641559.1](#) *Trichoderma viridarium* partial tef1a gene for translation elongation factor 1 alpha

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 868C..... 889

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1135C..A 1116

>[MT430084.1](#) *Claviceps purpurea* isolate LM642 translation elongation factor 1-alpha (TEF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 208C..... 229

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 475T..C 456

>[PV755871.1](#) *Trichoderma* sp. strain CFCC 88172A translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 807C..... 828

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1074C..G 1055

>[KU758964.1](#) *Trichoderma* sp. strain mms1255 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 801C..... 822

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1068Y..G 1049

>[ON783871.1](#) *Claviceps* sp. voucher DAOMC 250812 translation elongation factor 1-alpha (EF1-alpha) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 325C..... 346

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 592C..C 573

>[KF049694.1](#) *Polycephalomyces nipponicus* voucher BCC 1682 translation elongation factor 1-alpha gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 221G..... 242

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 488C..G 469

>[OQ944825.1](#) *Clonostachys setosa* culture CBS:112025 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 241A..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 508C..A 489

>[OQ470758.1](#) *Acremonium egyptiacum* isolate CBS113719 translation elongation factor 1-alpha

(tef1) gene, partial cds

product length = 271

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 241C..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 511C..G 492

>MK040114.1 Beauveria bassiana isolate B23 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 272T..... 293

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 539C..G 520

>MT430066.1 Claviceps purpurea isolate LM710 translation elongation factor 1-alpha (TEF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 208C..... 229

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 475T..C 456

>OQ470876.1 Emericellopsis minima isolate CBS190.55 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 241C..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 508C..G 489

>PV166505.1 Gibellula penicilliioides strain RCEF7797 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 263A..... 284

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 530C..A 511

>MN401471.1 Beauveria asiatica strain BCC2086 translation elongation factor 1 alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 319T..... 340

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 586C..G 567

>[DQ835478.1](#) *Trichoderma microcitrium* strain G.J.S. 91-61 translation elongation factor 1-alpha (tef) gene, exon and partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 259C..... 280

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 526C..G 507

>[MT039299.1](#) *Purpureocillium lilacinum* strain PLC3 translation elongation factor 1 alpha (tef1alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 295C..... 316

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 562C..G 543

>[KP008854.1](#) *Trichoderma virens* strain TRS106 translation elongation factor 1-alpha (tef1) gene, exons 5 and 6 and partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 742C..... 763

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 1009C..G 990

>[MT430035.1](#) *Claviceps purpurea* isolate LM14 translation elongation factor 1-alpha (TEF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 208C..... 229

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 475T..C 456

>[OQ471101.1](#) *Paraglimastix luzulae* isolate CBS494.67 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 241C..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 508C..G 489

>[KX066256.1](#) *Trichoderma* sp. 3 WYZ-2016 strain HMAS 248748 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 683C..... 704

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 950C..G 931

>[PV166523.1](#) *Gibellula penicilliioides* strain GNJ20240708-04 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 275A..... 296

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 542C..A 523

>[KY688038.1](#) *Trichoderma bannaense* voucher HMAS:248865 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 812C..... 833

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 1079C..G 1060

>[OP562779.1](#) *Trichoderma atroviride* isolate HF2P30 elongation factor 1-alpha (tef-1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 859C..... 880

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 1126C..A 1107

>[KU854950.1](#) *Ophiocordyceps formosana* voucher MFLU:15-3889 translation elongation factor 1-alpha (EF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 292C..... 313

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 559C..A 540

>[PP848447.1](#) *Trichoderma koningiopsis* strain ZHKUCC 24-0791 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 805C..... 826

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 1072C..G 1053

>[MT069951.1](#) *Trichoderma longibrachiatum* strain TLS6 translation elongation factor alpha 1

(Tef1) gene, partial cds

```

product length = 268
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template        783      .....C.....  804

Reverse primer  1      GCTGCTCGTGGTGCATNTCN  20
Template       1050      .....C..A  1031

```

>[MT430189.1](#) *Claviceps purpurea* isolate Clav20 translation elongation factor 1-alpha (TEF1-alpha) gene, partial cds

```

product length = 268
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template        208      .....C.....  229

Reverse primer  1      GCTGCTCGTGGTGCATNTCN  20
Template       475      .....T..C  456

```

>[ON567764.1](#) *Ophiocordyceps bifertilis* strain YHH 20162 translation elongation factor 1 alpha (tef) gene, partial cds

```

product length = 268
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template        304      .....C.....  325

Reverse primer  1      GCTGCTCGTGGTGCATNTCN  20
Template       571      .....C..A  552

```

>[MN605882.1](#) *Trichoderma vermifimicola* strain CGMCC 3.19694 translation elongation factor 1-alpha (tef1) gene, partial cds

```

product length = 268
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template        810      .....C.....  831

Reverse primer  1      GCTGCTCGTGGTGCATNTCN  20
Template       1077      .....C..G  1058

```

>[FJ860623.1](#) *Hypocrea margaretensis* strain C.P.K. 994 translation elongation factor 1 alpha (tef1) gene, partial sequence

```

product length = 268
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template        896      .....C.....  917

Reverse primer  1      GCTGCTCGTGGTGCATNTCN  20
Template       1163      .....C..G  1144

```

>[KT735256.1](#) *Trichoderma tibetense* voucher HMAS:248721 translation elongation factor 1-alpha (tef1) gene, partial cds

```

product length = 268
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template        807      .....C.....  828

Reverse primer  1      GCTGCTCGTGGTGCATNTCN  20

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Template 1074C..G 1055

>[OQ944532.1](#) *Sesquicillium phyllophilum* culture CBS:921.97 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 241T..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 508T..G 489

>[PV157964.1](#) *Trichoderma capillare* strain SLO1-1 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 1251C..... 1272

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1518C..G 1499

>[OR602795.1](#) *Akanthomyces muscarius* strain 12-Oct-AFF-FC5 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 301T..... 322

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 568C..G 549

>[MT027504.1](#) *Akanthomyces attenuatus* strain BBC9 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 295T..... 316

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 562C..G 543

>[MN523550.1](#) *Beauveria pseudobassiana* voucher YHH 1806005 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 328T..... 349

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 595C..G 576

>[PV346763.1](#) *Trichoderma* sp. WD-2025a isolate YMF1.09956 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 771T..... 792

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1038C..G 1019

>[PV755831.1](#) *Trichoderma* sp. strain CFCC 55234A translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 819C..... 840

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1086C..A 1067

>[AB758478.1](#) *Metacordyceps chlamydosporia* EF1-alpha gene for translation elongation factor 1 alpha, complete cds, strain: JCM 18607

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 296C..... 317

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 563C..G 544

>[MH986251.1](#) *Metarhizium brunneum* isolate AGF124 translation elongation factor 1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 233T..... 254

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 500T..C 481

>[AY391989.1](#) *Trichoderma sinuosum* strain G.J.S. 90-41 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 183C..... 204

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 450C..G 431

>[PV273527.1](#) *Emericellopsis fimetaria* strain CBS 548.84 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 241C..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 508C..G 489

>[OP620753.1](#) *Trichoderma koningii* isolate PL110114 translation elongation factor 1-alpha (EF1-

alpha) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 299C..... 320
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 566C..G 547

>[ON512608.1](#) Tolypocladium sp. isolate JMS367 translation elongation factor EF1-alpha (tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 244C..... 265
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 511C..C 492

>[OR667323.1](#) Trichoderma afroharzianum culture SZMC:25231 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 816C..... 837
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1083C..G 1064

>[OM906892.1](#) Trichoderma hamatum isolate SG3 translation elongation factor 1-alpha (TEF1a) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 832T..... 853
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1099C..G 1080

>[LC684173.1](#) Claviceps miscanthicola TNS:F-96455 TEF gene for translation elongation factor 1-alpha, partial sequence

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 326T..... 347
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 593T..C 574

>[EF468792.1](#) Paecilomyces lilacinus strain CBS 284.36 translation elongation factor 1 alpha-like (tef) gene, partial sequence

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 248C..... 269
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 515C..A 496

>[DQ345346.1](#) *Hypocrea crystalligena* translation elongation factor 1 alpha (tef1) gene, partial sequence

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 839T..... 860

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 1106C..G 1087

>[MW292448.1](#) *Ophiocordyceps asiana* isolate MY11878 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 256T..... 277

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 523T..A 504

>[EF469073.1](#) *Torrubiella wallacei* strain CBS 101237 translation elongation factor 1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 328T..... 349

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 595C..G 576

>[OQ471095.1](#) *Paracylindrocarpon multiseptatum* isolate CBS337.77 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 241C..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 508C..G 489

>[OP832376.1](#) *Simplicillium lanosoniveum* voucher CHE-CNRCB-977 translation elongation factor 1 alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 292A..... 313

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 559C..G 540

>[AY531907.1](#) *Beauveria bassiana* isolate 1969 translation elongation factor 1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 1039T..... 1060

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 1306C..G 1287

>[KR049184.1](#) *Metarhizium pemphigi* strain RCEF1003 translation elongation factor 1 alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 290C..... 311

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 557C..C 538

>[PV259760.1](#) *Stilbocrea gracilipes* strain UESTCC 01.0212 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 246C..... 267

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 513C..G 494

>[MW232544.1](#) *Trichoderma* sp. isolate T17-35 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 849C..... 870

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 1116C..G 1097

>[MH458405.1](#) *Beauveria pseudobassiana* strain YNHHCS4 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 302T..... 323

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 569C..G 550

>[MT430076.1](#) *Claviceps purpurea* isolate LM651 translation elongation factor 1-alpha (TEF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 208C..... 229

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 475T..C 456

>[OQ927366.1](#) *Beauveria bassiana* strain NRID68 translation elongation factor-1 alpha gene, partial

cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 268T..... 289

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 535C..G 516

>[OP313801.1](#) *Ophiocordyceps ovatospora* strain YHH 2206001 translation elongation factor 1 alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 295C..... 316

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 562C..G 543

>[MT429981.1](#) *Claviceps purpurea* isolate LM209 translation elongation factor 1-alpha (TEF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 208C..... 229

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 475T..C 456

>[OQ979225.1](#) *Hirsutella subulata* strain HKAS 126033 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 271

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 302C..... 323

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 572C..G 553

>[LT220758.1](#) *Beauveria bassiana* partial tef1a gene for translation elongation factor 1 alpha, strain TM1613

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 294C..... 315

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 561C..G 542

>[EU392641.1](#) *Hypocrella disciformis* strain P.C. 575 translation elongation factor 1 alpha (TEF1alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 296C..... 317

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 563T..G 544

>[MH521887.1](#) *Hevansia novoguineensis* strain BCC36270 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 263T..... 284

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 530C..G 511

>[KP276662.1](#) *Polyccephalomyces agaricus* strain YHCPA1307 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 306G..... 327

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 573C..G 554

>[PV273668.1](#) *Fusariella concinna* strain CBS 367.76 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 241C..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 508C..G 489

>[DQ672609.2](#) *Trichoderma olivascens* strain CBS 119322 translation elongation factor 1 alpha (tef1) gene, partial sequence

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 840C..... 861

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 1107C..A 1088

>[OP585415.1](#) *Hypocreales* sp. FY-2022a strain KUNCC21-10030 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 295T..... 316

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 562C..G 543

>[OQ470906.1](#) *Fusariella hughesii* isolate CBS435.70 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 241C..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 508C..G 489

>[PV617260.1](#) *Trichoderma* sp. strain CFCC 88437 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 369C..... 390

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 636C..G 617

>[MW245732.1](#) *Beauveria pseudobassiana* isolate BUB1135 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 301T..... 322

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 568C..G 549

>[PV107625.1](#) *Trichoderma* sp. isolate DG1143 translation elongation factor 1-alpha (TEF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 106C..... 127

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 373C..A 354

>[KX713677.1](#) *Ophiocordyceps camponoti-atricipis* strain ATRI3 translation elongation factor 1 alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 238C..... 259

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 505C..G 486

>[EU392646.1](#) *Hypocrella discoidea* strain I93-901D translation elongation factor 1 alpha (TEF1alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 288C..... 309

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 555C..G 536

>[GU233722.1](#) *Beauveria bassiana* strain RCEF0835 translation elongation factor 1 alpha gene,

partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 54T..... 75

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 321C..G 302

>[LC382182.1](#) *Parengyodontium album* UAMH 4512 tef1 gene for translation elongation factor 1-alpha, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 314C..... 335

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 581T..A 562

>[ON649930.1](#) *Trichoderma atroviride* isolate JZBQT8Z4 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 851C..... 872

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1118C..A 1099

>[MT430141.1](#) *Claviceps purpurea* isolate LM423 translation elongation factor 1-alpha (TEF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 208C..... 229

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 475T..C 456

>[PV166530.1](#) *Hevansia* sp. strain LS20230802-68 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 295T..... 316

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 562C..G 543

>[LC684216.1](#) *Claviceps paspali* TNS:F-96504 TEF gene for translation elongation factor 1-alpha, partial sequence

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 326C..... 347

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 593T..C 574

>[OQ471056.1](#) *Neoacremonium distortum* isolate CBS291.70A translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 241T..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 508C..G 489

>[JN715626.1](#) *Trichoderma caerulescens* strain S216 translation elongation factor 1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 820C..... 841

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1087C..G 1068

>[LC496904.1](#) *Simplicillium aogashimaense* JCM 18167 TEF1a gene for translation elongation factor 1 alpha, partial sequence

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 295A..... 316

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 562C..G 543

>[DQ069995.1](#) *Aschersonia aleyrodis* strain CUP67341 elongation factor 1-alpha-like gene, partial sequence

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 194T..... 215

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 461C..G 442

>[OR759136.1](#) *Hypomyces sichuanensis* translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 288C..... 309

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 555C..A 536

>[OP429591.1](#) *Paraisaria* sp. strain MY11264E translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 322C..... 343

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 589C..C 570

>[OK216930.1](#) *Akanthomyces dipterigenus* strain SCA81 translation elongation factor 1-alpha gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 198T..... 219

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 465C..G 446

>[LC684199.1](#) *Claviceps sorghicola* MAFF 306571 TEF gene for translation elongation factor 1-alpha, partial sequence

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 326T..... 347

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 593T..C 574

>[KJ878981.1](#) *Ophiocordyceps* sp. OSC 151909 translation elongation factor 1-alpha gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 277C..... 298

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 544C..C 525

>[OQ440687.1](#) *Hevansia novoguineensis* voucher KUMCC 8676 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 285T..... 306

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 552C..G 533

>[MH521885.1](#) *Hevansia novoguineensis* strain CBS 610.80 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 291T..... 312

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 558C..G 539

>[MK504359.1](#) *Beauveria pseudobassiana* strain NREP103 translation elongation factor 1-alpha

gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 206T..... 227

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 473C..G 454

>[ON101295.1](#) *Aciculosporium take* isolate SICAUCC 22-0017 translation elongation factor 1-alpha (tefA) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 307C..... 328

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 574T..A 555

>[KJ536058.1](#) *Beauveria pseudobassiana* strain EABb 09/04-Fil translation elongation factor 1-alpha (tef1a) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 321T..... 342

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 588C..G 569

>[MT249280.1](#) *Trichoderma harzianum* isolate 3 translation elongation factor 1-alpha (tef1) gene, exons 5 and 6 and partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 796C..... 817

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1063C..G 1044

>[ON649931.1](#) *Trichoderma atroviride* isolate JZBQT8Z5 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 860C..... 881

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1127C..A 1108

>[LC425536.1](#) *Parengyodontium album* CBS 121919 TEF1a gene for translation elongation factor 1a, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 317C..... 338

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 584T..A 565

>[KU974003.1](#) *Trichoderma hypoxylon* strain CGMCC 3.17907 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 302T..... 323

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 569C..G 550

>[OQ440706.1](#) *Conoideocrella luteorostrata* voucher KUMCC 9129 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 277T..... 298

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 544C..A 525

>[PP945769.1](#) *Gibellula solita* strain BCC_39935 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 321C..... 342

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 588C..G 569

>[OR855797.1](#) *Ophiocordyceps kohchangensis* strain BCC90811 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 292C..... 313

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 559C..C 540

>[AB968594.1](#) *Paraisaria heteropoda* EF1-alpha gene for translation elongation factor 1-alpha, partial cds, culture_collection: NBRC:100642

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 326C..... 347

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 593C..C 574

>[JN715641.1](#) *Trichoderma caerulescens* strain S434 translation elongation factor 1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 821C..... 842

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1088C..G 1069

>[MF371226.1](#) *Trichoderma* sp. strain TC714 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 759C..... 780

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1026C..A 1007

>[LC684220.1](#) *Claviceps paspali* TNS:F-96509 TEF gene for translation elongation factor 1-alpha, partial sequence

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 326C..... 347

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 593T..C 574

>[PV050422.1](#) *Gliomastix fasciculata* strain WL05659 from China translation elongation factor 1-alpha (tef1a) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 295C..... 316

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 562C..G 543

>[OR115233.1](#) *Cordycipitaceae* sp. QZW-2023a strain YFCC 863.01 translation elongation factor 1 alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 328T..... 349

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 595C..C 576

>[OR965915.1](#) *Metarhizium* sp. isolate MM2 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 351T..... 372

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 618T..C 599

>[KF747102.1](#) *Tolypocladium* RG-2013a strain MS460 translation elongation factor 1 alpha (tef)

gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 243C..... 264

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 510C..C 491

>[MK860196.1](#) *Beauveria bassiana* voucher GACP 16041002 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 313T..... 334

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 580C..G 561

>[KX914444.1](#) *Beauveria* sp. AI-2016a isolate 2931 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 305T..... 326

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 572C..G 553

>[OU720070.1](#) *Claviceps digitariae* partial tef1a gene for translation elongation factor 1 alpha, strain CCC1263

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 325C..... 346

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 592T..C 573

>[PP654214.1](#) *Pleurocordyceps puerensis* voucher GACPY117 translation elongation factor 1-alpha (EF1alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 292A..... 313

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 559C..G 540

>[EF193189.1](#) *Beauveria bassiana* isolate KVL 03_84 translation elongation factor 1 alpha gene, partial sequence

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 1039T..... 1060

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 1306C..G 1287

>[KJ364494.1](#) *Metacordyceps taii* voucher GZUH2012HK6 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 280T..... 301

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 547T..C 528

>[OQ944817.1](#) *Clonostachys samuelsii* culture CBS:188.94 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 241T..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 508C..A 489

>[MT131818.1](#) *Simplicillium lanosoniveum* voucher CER1-542 translation elongation factor 1-alpha (TEF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 319A..... 340

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 586C..G 567

>[KJ634757.1](#) *Trichoderma applanatum* isolate 7781 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 880C..... 901

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 1147C..G 1128

>[KJ665709.1](#) *Trichoderma simmonsii* strain S280 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 810C..... 831

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 1077C..G 1058

>[PV273480.1](#) *Hapsidospora globosa* strain CBS 268.91 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 217C..... 238

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 484C..G 465

>[KC285675.1](#) *Trichoderma paraviridescens* strain S32 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 836C..... 857

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1103C..A 1084

>[OQ944514.1](#) *Sesquicillium lasiacidis* culture CBS:179.88 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 241T..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 508C..A 489

>[KJ665513.1](#) *Trichoderma hamatum* strain Hypo 647 voucher WU:31629 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 816T..... 837

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1083C..G 1064

>[AF543777.1](#) *Epichloe typhina* strain ATCC 56429 translation elongation factor 1 alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 328T..... 349

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 595T..A 576

>[PV608810.1](#) *Alfaria cyperi-esculenti* strain XG308a translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 259A..... 280

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 526C..G 507

>[OM482397.1](#) *Purpureocillium atypicola* strain RCEF3837 translation elongation factor 1-alpha (tef)

gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 288C..... 309

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 555C..A 536

>[MH521888.1](#) *Jenniferia cinerea* strain BCC47914 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 262T..... 283

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 529C..G 510

>[KJ665565.1](#) *Trichoderma mediterraneum* strain S174 voucher WU:33329 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 837T..... 858

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1104C..G 1085

>[KJ665713.1](#) *Trichoderma simmonsii* strain S311 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 809C..... 830

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1076C..G 1057

>[KC285634.1](#) *Trichoderma viridescens* strain S1 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 836C..... 857

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1103C..A 1084

>[AY986939.1](#) *Aschersonia viridans* strain M.L. 202i translation elongation factor 1-alpha (EF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 295C..... 316

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 562T..G 543

>[KC708022.1](#) *Purpureocillium lilacinum* strain RRC 1815 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 292C..... 313

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 559C..A 540

>[OQ470834.1](#) *Cylindromonium lichenicola* isolate CBS188.70 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 271

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 241T..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 511C..A 492

>[PV617219.1](#) *Trichoderma* sp. strain CFCC 55234 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 711C..... 732

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 978C..A 959

>[MT070156.1](#) *Trichoderma achlamydosporum* strain YMF 1.06226 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 793C..... 814

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1060C..G 1041

>[FJ860659.1](#) *Trichoderma ochroleucum* strain CBS 119502 translation elongation factor 1 alpha (tef1) gene, partial sequence

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 864C..... 885

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1131C..G 1112

>[PV617257.1](#) *Trichoderma* sp. strain CFCC 88148 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 720C..... 741

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 987C..G 968

>[JQ257028.1](#) *Cordyceps fraticida* voucher TNS 19011 translation elongation factor 1 alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 277C..... 298

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 544C..A 525

>[KY688048.1](#) *Trichoderma linzhiense* voucher HMAS:248874 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 816C..... 837

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1083C..G 1064

>[OQ186370.1](#) *Beauveria medogensis* voucher HKAS:102452 translation elongation factor 1-alpha (tef1a) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 292T..... 313

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 559C..G 540

>[JX083709.1](#) *Claviceps humidiphila* isolate WAB1 translation elongation factor 1-alpha (EF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 325C..... 346

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 592T..C 573

>[OQ927367.1](#) *Beauveria bassiana* strain NRID69 translation elongation factor-1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 268T..... 289

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 535C..G 516

>[MK632068.1](#) *Beauveria gryllotalpidicola* strain BCC81472 translation elongation factor 1-alpha

(tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 316T..... 337

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 583C..G 564

>MF959741.1 Pleurocordyceps phaothaiensis strain BCC84557 translation elongation factor 1 alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 277A..... 298

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 544C..G 525

>PP855555.1 Matutiniella brasiliensis strain CDA 3362 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 292C..... 313

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 559C..C 540

>PQ044387.1 Clavicipitaceae sp. LZ-2024a isolate A5T-2-7 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 294T..... 315

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 561C..C 542

>KY087818.1 Metarhizium gaoligongense isolate BUM3.5 translation elongation factor 1-alpha (EF1alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 312C..... 333

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 579C..C 560

>OQ791275.1 Trichoderma citrinoviride strain Cin-9 translation elongation factor 1-alpha (tef1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 334C..... 355

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 601C..G 582

>[LC496905.1](#) *Simplicillium aogashimaense* JCM 18168 TEF1a gene for translation elongation factor 1 alpha, partial sequence

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 243A..... 264

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 510C..G 491

>[DQ384975.1](#) *Hypocrella discoidea* strain BCC 7865 translation elongation factor 1-alpha (EF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 292T..... 313

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 559T..G 540

>[KJ665678.1](#) *Trichoderma polysporum* strain S307 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 855T..... 876

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1122C..G 1103

>[OQ124002.1](#) *Trichoderma virens* strain JZB3360018 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 796C..... 817

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1063C..G 1044

>[HQ026480.1](#) *Claviceps purpurea* translation elongation factor 1-alpha (tefA) gene, complete cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 1114C..... 1135

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1381T..C 1362

>[LC496909.1](#) *Simplicillium minatense* JCM 18178 TEF1a gene for translation elongation factor 1 alpha, partial sequence

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 241A..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 508C..G 489

>[JF724132.1](#) *Myrothecium roridum* strain 781 translation elongation factor 1 alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 296C..... 317

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 563C..G 544

>[PQ048033.1](#) *Beauveria bassiana* isolate LH10 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 300T..... 321

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 567C..G 548

>[KC870067.1](#) *Metarhizium pinghaense* isolate RMT4 translation elongation factor 1-alpha (EF1alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 1011T..... 1032

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1278T..C 1259

>[MT430171.1](#) *Claviceps purpurea* isolate Clav46 translation elongation factor 1-alpha (TEF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 208C..... 229

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 475T..C 456

>[OQ944648.1](#) *Clonostachys levigata* culture CBS:101916 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 241T..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 508T..G 489

>[MW232535.1](#) *Trichoderma* sp. isolate T17-25 translation elongation factor 1-alpha (tef1) gene,

partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 791C..... 812

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1058C..G 1039

>[FJ765266.1](#) *Cordyceps* sp. BCC15603 translation elongation factor 1 alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 289T..... 310

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 556C..G 537

>[OL981629.1](#) *Gibellula longispora* strain GNJ20210711-22 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 295C..... 316

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 562C..G 543

>[PV617208.1](#) *Trichoderma* sp. strain CFCC 53491 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 721C..... 742

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 988C..G 969

>[PV368560.1](#) *Akanthomyces* sp. strain LV004 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 294T..... 315

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 561C..G 542

>[PV435855.1](#) *Simplicillium lanosoniveum* voucher CHE-CNRCB 1273 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 330A..... 351

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 597C..G 578

>[PQ671894.1](#) *Beauveria bassiana* voucher CHE-CNRCB 408 translation elongation factor (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 324T..... 345

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 591C..G 572

>[KX713681.1](#) *Ophiocordyceps ootakii* strain J13 translation elongation factor 1 alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 235C..... 256

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 502C..G 483

>[PP076819.1](#) *Memnoniella cunninghamiae* isolate UESTCC 23.0233 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 302C..... 323

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 569C..G 550

>[MT881872.1](#) *Trichoderma longibrachiatum* strain 18EPLE005 translation initiation factor 1 alpha (TEF1a) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 828C..... 849

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1095C..A 1076

>[OQ858773.1](#) *Trichoderma virens* isolate V078-S4-1 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 804C..... 825

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1071C..G 1052

>[PV617232.1](#) *Trichoderma* sp. strain CFCC 80922 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 690C..... 711

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 957C..G 938

>[PQ658982.1](#) *Beauveria bassiana* strain NR-OM26 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 250T..... 271

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 517C..G 498

>[PX071843.1](#) *Emericellopsis fuci* strain SFC20250301-M026 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 242C..... 263

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 509C..G 490

>[MN218253.1](#) *Ophiocordyceps unilateralis* isolate P.V1 translation elongation factor 1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 298C..... 319

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 565C..G 546

>[OR687154.1](#) *Trichoderma parareesei* strain N4-3 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 849C..... 870

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1116C..G 1097

>[MF125298.1](#) *Trichoderma harzianum* strain N-10 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 804C..... 825

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1071C..G 1052

>[PQ630182.1](#) *Ophiocordyceps* sp. NS-2024a isolate GZUHB01S translation elongation factor 1-

alpha (EF1alpha) gene, partial cds

```

product length = 268
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template        287    .....T.....  308

Reverse primer  1      GCTGCTCGTGGTGCATNTCN  20
Template        554    .....T..G  535

```

>[MG770895.1](#) *Trichoderma virens* strain B2115 translation elongation factor 1 alpha (TEF1alpha) gene, partial cds

```

product length = 268
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template        712    .....C.....  733

Reverse primer  1      GCTGCTCGTGGTGCATNTCN  20
Template        979    .....C..G  960

```

>[OQ471241.1](#) *Verruciconidia infusata* isolate CBS100888 translation elongation factor 1-alpha (tef1) gene, partial cds

```

product length = 268
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template        241    .....C.....  262

Reverse primer  1      GCTGCTCGTGGTGCATNTCN  20
Template        508    .....C..G  489

```

>[JN819021.1](#) *Ophiocordyceps unilateralis* isolate Co13 translation elongation factor 1 alpha (tef) gene, partial cds

```

product length = 268
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template        330    .....C.....  351

Reverse primer  1      GCTGCTCGTGGTGCATNTCN  20
Template        597    .....C..G  578

```

>[JN819001.1](#) *Hirsutella formicarum* isolate Po23 translation elongation factor 1 alpha (tef) gene, partial cds

```

product length = 268
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template        328    .....C.....  349

Reverse primer  1      GCTGCTCGTGGTGCATNTCN  20
Template        595    .....C..G  576

```

>[PV166536.1](#) *Purpureocillium atypicola* strain WLT20231225-1 translation elongation factor 1-alpha (TEF1) gene, partial cds

```

product length = 268
Forward primer  1      TGGTACAAGGGNTGGGAGAAGG  22
Template        297    .....C.....  318

Reverse primer  1      GCTGCTCGTGGTGCATNTCN  20

```

Template 564C..A 545

>[AY531916.1](#) *Beauveria bassiana* isolate 2579 translation elongation factor 1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 1048C..... 1069

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1315C..G 1296

>[EF469067.1](#) *Lecanicillium psalliotae* strain CBS 532.81 translation elongation factor 1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 328T..... 349

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 595C..A 576

>[MH070614.1](#) *Beauveria bassiana* isolate BinsAR2 elongation factor-1 alpha (EF1-a) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 279T..... 300

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 546C..G 527

>[OQ471196.1](#) *Simplicillium lanosoniveum* isolate CBS321.72 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 241A..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 508C..G 489

>[OQ954487.1](#) *Amphichorda coprophila* strain CBS 247.82 translation elongation factor 1-alpha (Tef1a) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 300C..... 321

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 567C..G 548

>[KY688051.1](#) *Trichoderma solum* voucher HMAS:248849 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 817C..... 838

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1084C..G 1065

>[PQ569876.1](#) *Metarhizium* sp. YY-2024a voucher HKAS 132273 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 246C..... 267

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 513T..C 494

>[MT070148.1](#) *Trichoderma subazureum* strain YMF 1.06207 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 771C..... 792

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1038C..G 1019

>[MT701573.1](#) *Metapochonia bulbillosa* isolate P6620 translation elongation factor 1-alpha (tef 1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 301T..... 322

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 568C..A 549

>[MW245722.1](#) *Metarhizium brunneum* isolate BUM1353 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 301T..... 322

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 568C..G 549

>[MT070153.1](#) *Trichoderma simile* strain YMF 1.6180 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 696C..... 717

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 963C..G 944

>[OQ471073.1](#) *Nothoacremonium exiguum* isolate CBS587.73 translation elongation factor 1-alpha

(tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 241C..... 262
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 508C..A 489

>[KX066255.1](#) *Trichoderma* sp. 3 WYZ-2016 strain HMAS 273788 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 842C..... 863
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1109C..G 1090

>[LC684232.1](#) *Claviceps queenslandica* TNS:F-96522 TEF gene for translation elongation factor 1-alpha, partial sequence

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 326C..... 347
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 593T..C 574

>[OL978579.1](#) *Lecanicillium* sp. isolate HK translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 302T..... 323
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 569C..G 550

>[OQ470880.1](#) *Emericellopsis moniliformis* isolate CBS463.92 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 241C..... 262
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 508C..G 489

>[KM510182.1](#) *Metarhizium brunneum* strain RCEF5573 translation elongation factor 1 alpha (TEF) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 287T..... 308
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 554T..C 535

>[OL606738.1](#) *Alfaria cyperi-esculenti* voucher MFLU 16-1344 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 261A..... 282

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 528C..G 509

>[OR762226.1](#) *Trichoderma* sp. JS-2023b strain URMicro 11721 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 913C..... 934

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1180C..G 1161

>[PQ758609.1](#) *Sesquicillium buxi* strain GMBC 3000 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 313T..... 334

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 580C..A 561

>[KY688044.1](#) *Trichoderma longisporum* voucher HMAS:248868 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 820C..... 841

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1087C..G 1068

>[PP780915.1](#) *Beauveria malawiensis* isolate KUNCC 8904 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 1036T..... 1057

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1303C..G 1284

>[MT069946.1](#) *Trichoderma afroharzianum* strain TAS8 translation elongation factor alpha 1 (Tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 812C..... 833

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1079C..G 1060

>[KJ665524.1](#) *Trichoderma italicum* strain S128 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 810C..... 831

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1077C..A 1058

>[KC285690.1](#) *Trichoderma virilente* strain S108 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 842C..... 863

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1109C..G 1090

>[ON512610.1](#) *Tolypocladium* sp. isolate JMS369 translation elongation factor EF1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 244C..... 265

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 511C..C 492

>[KC998997.1](#) *Emericellopsis maritima* strain CBS 491.71 translation elongation factor 1-alpha (TEF1a) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 290C..... 311

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 557C..G 538

>[PV617197.1](#) *Trichoderma* sp. strain CFCC 53257 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 714C..... 735

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 981C..A 962

>[PV755880.1](#) *Trichoderma* sp. strain CFCC 89491A translation elongation factor 1-alpha (tef1)

gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 807T..... 828

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1074C..G 1055

>[KM225695.1](#) *Thyronectria rhodochlora* strain NP9 translation elongation factor 1-alpha (TEF1)
 gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 794T..... 815

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1061C..G 1042

>[MT430020.1](#) *Claviceps purpurea* isolate LM37 translation elongation factor 1-alpha (TEF1-alpha)
 gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 208C..... 229

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 475T..C 456

>[MW292444.1](#) *Ophiocordyceps asiana* isolate BCC86879 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 277T..... 298

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 544T..G 525

>[AF534622.1](#) *Trichoderma longipile* strain DAOM 177227 translation elongation factor 1-alpha (ef1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 183C..... 204

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 450C..G 431

>[PP658210.1](#) *Purpureocillium jiangxiense* isolate JX13B01 translation elongation factor 1-alpha
 gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 231C..... 252

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 498C..A 479

>[PV184573.1](#) *Trichoderma guizhouense* strain TRPN17 translation elongation factor 1 (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 349C..... 370

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 616C..G 597

>[KM091841.1](#) *Isaria fumosorosea* strain GKVK 01_32 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 296T..... 317

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 563C..G 544

>[KJ878962.1](#) *Ophiocordyceps annulata* voucher CEM303 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 199C..... 220

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 466C..A 447

>[MF416467.1](#) *Cordyceps albocitrinus* voucher spat 07-174 translation elongation factor 1 alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 278T..... 299

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 545C..G 526

>[AB972951.1](#) *Polycephalomyces cuboideus* EF1-alpha gene for translation elongation factor 1-alpha, partial cds, culture_collection: NBRC:103836

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 325C..... 346

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 592C..C 573

>[OQ858739.1](#) *Trichoderma guizhouense* isolate IV72-S3-3 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 794C..... 815

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1061C..G 1042

>[MW723113.1](#) *Metarhizium* sp. strain RCEF6696 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 296T..... 317

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 563T..C 544

>[DQ835471.1](#) *Trichoderma sulphureum* strain G.J.S. 95-140 translation elongation factor 1-alpha (tef) gene, exon and partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 259C..... 280

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 526C..C 507

>[OR515612.1](#) *Metarhizium anisopliae* isolate 10005-2 translation elongation factor 1-alpha (EF1alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 299T..... 320

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 566T..C 547

>[MH746778.1](#) *Trichoderma samuelsii* strain 10837 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 828C..... 849

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1095C..A 1076

>[OM967099.1](#) *Trichoderma atroviride* strain LZ024 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 853C..... 874

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1120C..A 1101

>[XM_024548974.1](#) *Trichoderma gamsii* elongation factor 1-alpha (TGAM01_v202318), partial

mRNA

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 595C..... 616

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 862C..A 843

>[MT429982.1](#) *Claviceps purpurea* isolate LM215 translation elongation factor 1-alpha (TEF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 208C..... 229

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 475T..C 456

>[PV618248.1](#) *Beauveria bassiana* voucher CHE-CNRCB 1190 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 293T..... 314

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 560C..G 541

>[PQ166336.1](#) *Metarhizium* sp. strain URM 6108 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 281T..... 302

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 548T..C 529

>[OR525304.1](#) *Trichonectria vinosa* voucher KRAM-L-74382 translation elongation factor 1-alpha (TEF1a) gene, partial cds

product length = 271

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 273C..... 294

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 543C..G 524

>[ON101300.1](#) *Aciculosporium take* isolate SICAUCC 22-0022 translation elongation factor 1-alpha (tefA) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 306C..... 327

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 573T..A 554

>[PV617187.1](#) *Trichoderma* sp. strain CFCC 53055 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 690C..... 711

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 957C..G 938

>[MT081435.1](#) *Trichoderma koningiopsis* isolate Tkois1 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 840C..... 861

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1107C..G 1088

>[KR264016.1](#) *Hirsutella rhossiliensis* strain ARSEF3744 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 232C..... 253

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 499C..A 480

>[OR667305.1](#) *Trichoderma* sp. culture SZMC:12408 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 816C..... 837

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1083C..G 1064

>[ON512600.1](#) *Tolypocladium* sp. isolate JMS207 translation elongation factor EF1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 244C..... 265

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 511C..C 492

>[OQ440726.1](#) *Ophiocordyceps rhizoidea* voucher KUMCC 5211 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 285C..... 306

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 552C..G 533

>[ON934339.1](#) *Trichoderma afroharzianum* isolate a90 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 804C..... 825

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1071C..G 1052

>[PV755874.1](#) *Trichoderma* sp. strain CFCC 88574A translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 807T..... 828

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1074C..G 1055

>[PV568052.1](#) *Verruciconidia persicina* isolate MCC 9995 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 234C..... 255

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 501C..G 482

>[MW267335.1](#) *Trichoderma harzianum* isolate TA2-27 translation elongation factor 1-alpha (TEF-1 alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 789C..... 810

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1056C..G 1037

>[ON649893.1](#) *Trichoderma* sp. WTQ-2022g isolate JZBQT1Z2 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 812C..... 833

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1079C..G 1060

>[MN401497.1](#) *Beauveria bassiana* strain BCC79681 translation elongation factor 1 alpha (tef)

gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 319T..... 340

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 586C..G 567

>[OQ559120.1](#) *Trichoderma variegatum* strain YMF 1.7532 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 753T..... 774

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1020C..G 1001

>[ON512591.1](#) *Tolypocladium* sp. isolate JMS196 translation elongation factor EF1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 244C..... 265

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 511C..C 492

>[LC370972.1](#) *Ophiocordyceps* sp. gene for translation elongation factor 1-alpha, partial cds, strain: MiwITN01_OP

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 235C..... 256

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 502C..G 483

>[MF374811.1](#) *Trichoderma fujianense* strain 230 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 852C..... 873

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1119C..G 1100

>[KY688024.1](#) *Trichoderma pseudodensum* voucher HMAS:248829 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 813C..... 834

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 1080C..A 1061

>[KY415593.1](#) *Ophiocordyceps myrmicarum* voucher GZUIFR hirjc translation elongation factor 1-alpha (EF-1a) gene, partial cds

product length = 271

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 340C..... 361

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 610C..G 591

>[OL790432.1](#) *Trichoderma atroviride* isolate NECC21247 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 798C..... 819

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1065C..A 1046

>[OQ470805.1](#) *Caespitomonium squamicola* isolate CBS392.73 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 241T..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 508C..G 489

>[OU720044.1](#) *Claviceps cynodontis* partial tef1a gene for translation elongation factor 1 alpha, strain CCC1469

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 324C..... 345

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 591T..C 572

>[PP945752.1](#) *Gibellula longispora* strain BCC_21468 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 286C..... 307

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 553C..G 534

>[PV657660.1](#) *Trichoderma longibrachiatum* translation elongation factor 1-alpha (TEF1a) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 793C..... 814

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1060C..A 1041

>[PX071858.1](#) *Purpureocillium* sp. strain SFC20250301-M062 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 292C..... 313

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 559C..A 540

>[OQ980403.1](#) *Metarhizium putuoense* isolate HMAS285457 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 289C..... 310

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 556T..C 537

>[PP776174.1](#) *Moelleriella longzhuensis* strain YHH MLFSL2310013 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 305C..... 326

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 572T..G 553

>[OR501868.1](#) *Beauveria bassiana* isolate CEP 042 translation elongation factor 1a (EF1a) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 154T..... 175

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 421C..G 402

>[JX192798.1](#) *Hevansia novoguineensis* strain EPF037 translation elongation factor 1 alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 295T..... 316

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 562C..G 543

>[EF468749.1](#) *Ophiocordyceps entomorrhiza* strain KEW 53484 translation elongation factor 1 alpha

(tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 328C..... 349

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 595C..G 576

>MK284269.1 Ophiocordyceps khokpasiensis strain BCC 48071 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 277C..... 298

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 544C..G 525

>KP008860.1 Trichoderma virens strain TRS112 translation elongation factor 1-alpha (tef1) gene, exons 5 and 6 and partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 743C..... 764

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1010C..G 991

>JQ425683.1 Isaria javanica strain CBS 134.22 translation elongation factor 1 alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 286T..... 307

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 553C..G 534

>MG760729.1 Trichoderma koningiopsis strain B1896 translation elongation factor 1 alpha (TEF1alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 791C..... 812

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1058C..G 1039

>ON101288.1 Aciculosporium take isolate SICAUCC 22-0010 translation elongation factor 1-alpha (tefA) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 302C..... 323

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 569T..A 550

>[OQ186376.1](#) *Cordyceps cicadae* voucher HKAS:102460 translation elongation factor 1-alpha (tef1a) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 301T..... 322

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 568C..G 549

>[JQ867157.1](#) *Beauveria bassiana* strain RCEF0301 elongation factor-1 alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 305T..... 326

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 572C..G 553

>[MW690643.1](#) *Fusarium cf. oxysporum* isolate T745B translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 265

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 364T..... 385

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 628C..G 609

>[MZ514914.1](#) *Ophiocordyceps indica* isolate FB_7 translation elongation factor 1-alpha (EF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 161C..... 182

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 428C..A 409

>[LT903693.1](#) *Beauveria bassiana* genomic DNA containing Elongation factor region

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 297T..... 318

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 564C..G 545

>[PQ356325.1](#) *Bulbithecium* sp. strain URM9151 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 291C..... 312

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 558C..G 539

>[ON125026.1](#) *Niveomyces insectorum* isolate CBS 756.73 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 283C..... 304

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 550C..G 531

>[MH727390.1](#) *Ophiocordyceps sporangifera* culture MFLUCC:18-0492 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 291C..... 312

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 558C..G 539

>[PV274286.1](#) *Ophiocordyceps thilosuensis* strain BCC46607 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 295A..... 316

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 562T..G 543

>[HQ165688.1](#) *Nomuraea rileyi* strain NBRC8560 translation elongation factor 1 alpha (tef) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 292C..... 313

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 559T..C 540

>[PV855330.1](#) *Clonostachys rosea* isolate XX-6-2 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 295T..... 316

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 562C..A 543

>[KJ665419.1](#) *Trichoderma atroviride* strain S360 voucher WU:33363 translation elongation factor

1-alpha (tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 822C..... 843
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1089C..A 1070

>MH697676.1 Ophiocordyceps gracilis voucher YHH OG201305 translation elongation factor 1-alpha gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 302C..... 323
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 569C..C 550

>PV157963.1 Trichoderma sp. PAR10 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 1203C..... 1224
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1470C..G 1451

>MW168228.1 Beauveria felina strain YFCC 851 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 328C..... 349
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 595C..G 576

>MK040107.1 Beauveria bassiana isolate B13 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 272T..... 293
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 539C..G 520

>OQ054475.1 Hevansia mainiae strain BRIP 62571a translation elongation factor 1-alpha gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 295T..... 316
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 562C..G 543

>[MT430089.1](#) *Claviceps purpurea* isolate LM637 translation elongation factor 1-alpha (TEF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 208C..... 229

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 475T..C 456

>[MT430118.1](#) *Claviceps purpurea* isolate LM520 translation elongation factor 1-alpha (TEF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 208C..... 229

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 475T..C 456

>[MW267359.1](#) *Trichoderma harzianum* isolate TRA1-49 translation elongation factor 1-alpha (TEF-1 alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 786C..... 807

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1053C..G 1034

>[KJ665682.1](#) *Trichoderma polysporum* strain S458 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 853T..... 874

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1120C..G 1101

>[MH203473.1](#) *Beauveria bassiana* strain CHE-CNRCB 84 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 346C..... 367

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 613C..G 594

>[LC382184.1](#) *Parengyodontium album* UAMH 9836 tef1 gene for translation elongation factor 1-alpha, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 314C..... 335

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 581C..A 562

>[DQ070016.1](#) *Aschersonia placenta* strain AFR114 elongation factor 1-alpha-like gene, partial sequence

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 187T..... 208

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 454C..G 435

>[KC998975.1](#) *Emericellopsis alkalina* strain A116 translation elongation factor 1-alpha (TEF1a) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 290T..... 311

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 557C..A 538

>[KJ665389.1](#) *Trichoderma atrobrunneum* strain S392 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 817C..... 838

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 1084C..G 1065

>[OQ470932.1](#) *Acremonium polychromum* isolate CBS296.70B translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 241T..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 508C..G 489

>[OQ471199.1](#) *Stanjemonium grisellum* isolate CBS655.79 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 241T..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 508C..G 489

>[MN576953.1](#) *Liangia sinensis* strain YFCC 3104 translation elongation factor 1-alpha (TEF) gene,

partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 316T..... 337

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 583C..G 564

>[GQ250042.1](#) Cordyceps sp. BCC28585 translation elongation factor 1 alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 225T..... 246

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 492C..G 473

>[LC370801.1](#) Ophiocordyceps sp. gene for translation elongation factor 1-alpha, partial cds, strain: CfaKSD05_OP

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 235C..... 256

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 502C..G 483

>[OQ186365.1](#) Hypocrella raciborskii voucher MFLU:22-0261 translation elongation factor 1-alpha (tef1a) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 274T..... 295

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 541C..G 522

>[ON479090.1](#) Metarhizium pinghaense isolate M.pinghaense_EF1_923bp translation elongation factor 1-alpha (TEF) mRNA, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 296T..... 317

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 563T..C 544

>[OR908785.1](#) Pseudodiploospora longispora isolate GUCCX-091 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 278C..... 299

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 545C..G 526

>[OQ982010.1](#) *Pleurocordyceps* sp. YY-2023b strain GZCC 22-2041 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 293A..... 314

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 560C..G 541

>[MH397464.1](#) *Claviceps spartinae* voucher RUTPP:3419 translation elongation factor 1-alpha (Ef1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 235C..... 256

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 502T..C 483

>[OQ471217.1](#) *Trichothecium crotocinigenum* isolate CBS129.64 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 241T..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 508C..G 489

>[ON567754.1](#) *Ophiocordyceps subtiliphialida* strain YFCC 8814 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 295C..... 316

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 562C..G 543

>[KJ665396.1](#) *Trichoderma atrobrunneum* strain S447 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 817C..... 838

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1084C..G 1065

>[OQ944693.1](#) *Clonostachys ralfsii* culture CBS:141089 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 241T..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 508C..A 489

>[PP848432.1](#) *Trichoderma* sp. LT-2024d strain ZHKUCC 24-0776 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 812C..... 833

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1079C..G 1060

>[PQ516496.1](#) *Pseudodiploospora* sp. isolate BS-6 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 270C..... 291

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 537C..G 518

>[EF193193.1](#) *Beauveria bassiana* isolate KVL 03_92 translation elongation factor 1 alpha gene, partial sequence

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 1005T..... 1026

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1272C..G 1253

>[OQ944647.1](#) *Clonostachys kunmingensis* culture CBS:101920 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 241T..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 508C..A 489

>[MN218246.1](#) *Ophiocordyceps unilateralis* isolate P.M3 translation elongation factor 1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 298C..... 319

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 565C..G 546

>[MT906153.1](#) *Metarhizium robertsii* strain CHE-CNRCB 257 translation elongation factor 1 alpha

(EF1alpha) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 290T..... 311
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 557T..C 538

>[KJ665644.1](#) *Trichoderma petersenii* strain S325 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 817C..... 838
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1084C..G 1065

>[ON649908.1](#) *Trichoderma* sp. WTQ-2022e isolate JZBQT7Z10 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 816C..... 837
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1083C..G 1064

>[MK775508.1](#) *Trichoderma paraviride* strain YMF 1.04628 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 809C..... 830
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1076C..G 1057

>[LC163554.1](#) *Beauveria bassiana* EF1-a gene for translation elongation factor 1 alpha, partial cds, isolate: F794

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 326T..... 347
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 593C..G 574

>[AF534583.1](#) *Trichoderma piluliferum* strain CBS 814.68 translation elongation factor 1-alpha (ef1-alpha) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 183T..... 204
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 450C..G 431

>[PV166508.1](#) *Hevansia* sp. strain RCEF7801 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 274T..... 295

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 541C..G 522

>[MT430112.1](#) *Claviceps purpurea* isolate LM532 translation elongation factor 1-alpha (TEF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 208C..... 229

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 475T..C 456

>[AB968583.1](#) *Ophiocordyceps coccidiicola* EF1-alpha gene for translation elongation factor 1-alpha, partial cds, culture_collection: NBRC:100682

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 326C..... 347

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 593C..A 574

>[PV107602.1](#) *Simplicillium* sp. isolate NJB91 translation elongation factor 1-alpha (TEF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 151A..... 172

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 418C..A 399

>[KJ665640.1](#) *Trichoderma petersenii* strain S211 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 814C..... 835

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1081C..G 1062

>[MF416513.1](#) *Ophiocordyceps kniphofioides* voucher HUA TSF 790 translation elongation factor 1 alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 48C..... 69

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 315C..G 296

>[OQ944510.1](#) *Clonostachys candelabrum* strain JW79008 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 241T..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 508C..G 489

>[OQ944481.1](#) *Mycocitrus synnematus* culture CBS:126677 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 241T..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 508C..G 489

>[OQ858770.1](#) *Trichoderma virens* isolate SM109-S3-2 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 815C..... 836

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1082C..G 1063

>[LT220809.1](#) *Purpureocillium lilacinum* partial tef1a gene for translation elongation factor 1 alpha, strain GMCR57A

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 293C..... 314

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 560C..A 541

>[GQ250031.1](#) *Akanthomyces koratensis* strain BCC01485 translation elongation factor 1 alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 295T..... 316

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 562C..G 543

>[MT430081.1](#) *Claviceps purpurea* isolate LM646 translation elongation factor 1-alpha (TEF1-alpha)

gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 208C..... 229

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 475T..C 456

>[OR946326.1](#) *Ophiocordyceps fusiformis*pora strain YFCC 9014 translation elongation factor 1-alpha (tef1a) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 295T..... 316

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 562C..G 543

>[OQ067559.1](#) *Beauveria bassiana* isolate BLm1Rv19 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 319T..... 340

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 586C..G 567

>[OQ944541.1](#) *Clonostachys apocyni* culture CBS:130.87 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 241T..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 508C..A 489

>[PV755795.1](#) *Trichoderma* sp. strain CFCC 51938A translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 811C..... 832

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1078C..G 1059

>[PV166514.1](#) *Hevansia* sp. strain RCEF7806 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 272T..... 293

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 539C..G 520

>[KM283822.1](#) *Lecanicillium pissodis* strain CBS 118231 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 289T..... 310

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 556C..G 537

>[PP860724.1](#) *Trichoderma velutinum* isolate TrVe29 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 798C..... 819

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1065C..G 1046

>[KJ665621.1](#) *Trichoderma neorufoides* strain S306 voucher WU:33352 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 860C..... 881

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1127C..G 1108

>[MW507528.1](#) *Tolypocladium inusitaticapitatum* voucher HKAS 112153 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 296C..... 317

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 563T..C 544

>[OQ943208.1](#) *Beauveria polyrhachicola* voucher KUNCC 8636 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 322T..... 343

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 589C..G 570

>[MW723125.1](#) *Metarhizium viridulum* strain RCEF6804 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 199T..... 220

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 466C..C 447

>[OR115171.1](#) *Pochonia chlamydosporia* strain YFCC 802 translation elongation factor 1 alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 328C..... 349

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 595C..G 576

>[OP562781.1](#) *Trichoderma atroviride* isolate HF2P32 elongation factor 1-alpha (tef-1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 859C..... 880

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 1126C..A 1107

>[OQ944644.1](#) *Clonostachys intermedia* culture CBS:508.82 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 241T..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 508C..A 489

>[MH203475.1](#) *Beauveria bassiana* strain CHE-CNRCB 106 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 346T..... 367

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 613C..G 594

>[MH203477.1](#) *Beauveria bassiana* strain CHE-CNRCB 118 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 346T..... 367

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 613C..G 594

>[EU392666.1](#) *Moelleriella madidiensis* strain P.C. 594 translation elongation factor 1 alpha

(TEF1alpha) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 298C..... 319
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 565T..G 546

>[MW723121.1](#) *Metarhizium viridulum* strain RCEF6800 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 296T..... 317
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 563C..C 544

>[MT881880.1](#) *Trichoderma longibrachiatum* strain 18ESMA017 translation initiation factor 1 alpha (TEF1a) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 827C..... 848
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1094C..A 1075

>[OQ858752.1](#) *Trichoderma virens* isolate BN4-S1-1 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 821C..... 842
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1088C..G 1069

>[MN401502.1](#) *Beauveria asiatica* strain BCC14510 translation elongation factor 1 alpha (tef) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 319T..... 340
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 586C..G 567

>[PP464678.1](#) *Akanthomyces* sp. strain CGMCC3.25673 translation elongation factor 1-alpha (Tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 286T..... 307
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 553C..G 534

>[OM417522.1](#) *Akanthomyces kanyawimiae* translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 307C..... 328

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 574C..G 555

>[MW267423.1](#) *Trichoderma harzianum* isolate TA3-41 translation elongation factor 1-alpha (TEF-1 alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 789C..... 810

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1056C..G 1037

>[KJ665429.1](#) *Trichoderma aurantioeffusum* strain S48 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 890T..... 911

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1157C..G 1138

>[OR046018.1](#) *Trichoderma asperellum* isolate XT11 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 879T..... 900

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1146C..G 1127

>[OP562776.1](#) *Trichoderma atrobrunneum* isolate HF2P15 elongation factor 1-alpha (tef-1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 847C..... 868

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1114C..G 1095

>[PP059630.1](#) *Cordyceps cateniannulata* strain COAD:3680 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 268T..... 289

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 535C..G 516

>[OQ067535.1](#) *Beauveria bassiana* isolate BCI11As20 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 320C..... 341

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 587C..G 568

>[PP945773.1](#) *Gibellula* sp. DT-2024f strain BCC_30395 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 286C..... 307

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 553C..G 534

>[OQ471119.1](#) *Phialoseptomonium eucalypti* isolate CBS145542 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 241T..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 508C..G 489

>[MF416481.1](#) *Cordyceps* sp. voucher RCEF HP090724-04C translation elongation factor 1 alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 323T..... 344

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 590C..G 571

>[MN576998.1](#) *Beauveria pseudobassiana* strain YFCC 7120 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 302T..... 323

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 569C..G 550

>[EU369011.1](#) *Akanthomyces novoguineensis* strain NHJ 13161 translation elongation factor 1

alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 328T..... 349

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 595C..G 576

>[KC998996.1](#) *Emericellopsis minima* strain CBS 871.68 translation elongation factor 1-alpha (TEF1a) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 290C..... 311

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 557C..A 538

>[OR501876.1](#) *Beauveria bassiana* isolate CEP 322 translation elongation factor 1a (EF1a) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 154T..... 175

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 421C..G 402

>[PV617256.1](#) *Trichoderma* sp. strain CFCC 88146 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 722C..... 743

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 989C..G 970

>[OL322038.1](#) *Ascopolyporus griseoperithecatus* strain BCC25788 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 286C..... 307

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 553C..G 534

>[DQ835492.1](#) *Trichoderma subsulphureum* strain M 141 (Japan) translation elongation factor 1-alpha (tef) gene, exon and partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 259C..... 280

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 526C..A 507

>[PV012662.1](#) *Trichoderma harzianum* isolate TrHa167 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 799C..... 820

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1066C..G 1047

>[OQ944530.1](#) *Sesquicillium neerlandicum* culture CBS:148215 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 241T..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 508C..A 489

>[KC633274.1](#) *Akanthomyces sabanensis* strain JCh041 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 347T..... 368

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 614C..G 595

>[LT220776.1](#) *Clonostachys rosea* f. *rosea* partial tef1a gene for translation elongation factor 1 alpha, strain TMCR19

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 292T..... 313

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 559C..A 540

>[LC684121.1](#) *Claviceps purpurea* TNS:F-96398 TEF gene for translation elongation factor 1-alpha, partial sequence

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 326C..... 347

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 593T..C 574

>[ON934354.1](#) *Trichoderma obovatum* isolate E94 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 818C..... 839

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1085C..A 1066

>[OR199821.1](#) *Paraisaria* sp. 1 EB-2023 voucher OSC-M-052022 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 261C..... 282

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 528C..C 509

>[KR049187.1](#) *Metarhizium pemphigi* strain RCEF1989 translation elongation factor 1 alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 290C..... 311

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 557C..C 538

>[KU529133.1](#) *Trichoderma sphaerosporum* strain 9439 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 843C..... 864

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1110C..G 1091

>[KM504954.1](#) *Metarhizium anisopliae* strain RCEF3288 translation elongation factor 1 alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 268T..... 289

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 535T..C 516

>[KC870069.1](#) *Metarhizium pinghaense* isolate RMT16 translation elongation factor 1-alpha (EF1alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 1006T..... 1027

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1273T..C 1254

>[OQ471253.1](#) *Verruciconidia persicina* isolate CBS439.66 translation elongation factor 1-alpha

(tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 241C..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 508C..G 489

>LT220748.1 Beauveria bassiana partial tef1a gene for translation elongation factor 1 alpha, strain GMCR67

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 294T..... 315

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 561C..G 542

>LN810537.1 Acremonium hyalinulum partial tef1a gene for translation elongation factor 1 alpha, strain CBS 560.86

product length = 271

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 299A..... 320

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 569C..G 550

>KF811547.1 Epichloe baconii strain 9707 translation elongation factor 1-alpha (tefA) gene, complete cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 1118T..... 1139

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1385T..A 1366

>OQ440743.1 Ophiocordyceps myrmecophila voucher HKAS 126052 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 285T..... 306

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 552T..G 533

>OQ858793.1 Trichoderma asperellum isolate PH1-S5-2 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 821T..... 842

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 1088C..G 1069

>[OR525312.1](#) *Trichonectria* sp. voucher KRAM-L-74385 translation elongation factor 1-alpha (TEF1a) gene, partial cds

product length = 271

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 316C..... 337

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 586C..G 567

>[MH244896.1](#) *Beauveria bassiana* strain AMEF110 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 287T..... 308

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 554C..G 535

>[PV273673.1](#) *Ijuhya corynospora* strain CBS 342.77 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 241C..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 508C..G 489

>[MW325731.1](#) *Trichoderma virens* isolate TRA3-25 translation elongation factor 1-alpha (TEF-1 alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 793C..... 814

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1060C..G 1041

>[PP059625.1](#) *Cordyceps cateniannulata* strain COAD:3675 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 268T..... 289

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 535C..G 516

>[OQ470795.1](#) *Bulbithecium borodinense* isolate CBS101148 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 241C..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 508C..G 489

>[ON815110.1](#) *Cordyceps* sp. isolate B6 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 334T..... 355

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 601C..G 582

>[OQ471004.1](#) *Lasionectria* sp. isolate CBS102040 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 241T..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 508C..G 489

>[OK392627.1](#) *Torrubiella arachnophila* isolate EBSL11 elongation factor 1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 290T..... 311

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 557C..G 538

>[DQ835479.1](#) *Trichoderma microcitrium* strain G.J.S. 97-248 translation elongation factor 1-alpha (tef) gene, exon and partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 259C..... 280

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 526C..G 507

>[OQ067575.1](#) *Beauveria bassiana* isolate BLm5KK21 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 317C..... 338

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 584C..G 565

>[OR115206.1](#) *Purpureomyces* sp. QZW-2023a strain YFCC 839 translation elongation factor 1 alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 328C..... 349

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 595C..C 576

>[OQ471274.1](#) *Waltergamsia fusidioides* isolate CBS705.86 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 271

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 241C..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 511C..G 492

>[PQ202113.1](#) *Ophiocordyceps* sp. isolate 67Leptaka_OP translation elongation factor 1-alpha (EF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 292C..... 313

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 559C..G 540

>[PP860725.1](#) *Trichoderma simmonsii* isolate TrSi23 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 804C..... 825

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1071C..G 1052

>[LT220791.1](#) *Cordyceps cicadae* partial tef1a gene for translation elongation factor 1 alpha, strain GMSL29

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 305T..... 326

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 572C..G 553

>[LC572039.1](#) *Aciculosporium sasicola* TNS:F91306 tef1a gene for translation elongation factor 1 alpha, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 325C..... 346

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 592T..A 573

>[OL322034.1](#) *Ascopolyporus purpuratus* strain BCC88389 translation elongation factor 1-alpha

gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 289C..... 310

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 556C..G 537

>[AY391977.1](#) *Trichoderma cinnamomeum* strain G.J.S. 96-128 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 183C..... 204

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 450C..G 431

>[MK504339.1](#) *Beauveria bassiana* strain AMEP052 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 247T..... 268

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 514C..G 495

>[OR188673.1](#) *Beauveria pseudobassiana* isolate NRXG04 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 274T..... 295

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 541C..G 522

>[KJ398813.1](#) *Metarhizium pemphigi* voucher ARSEF 6569 translation elongation factor 1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 1050C..... 1071

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1317C..C 1298

>[KJ634768.1](#) *Trichoderma britdaniae* isolate 8020 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 871T..... 892

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 1138C..G 1119

>[OR548062.1](#) *Trichoderma pararogersonii* isolate 219 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 836C..... 857

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1103C..G 1084

>[MZ190329.1](#) *Simplicillium minatense* isolate B2132D elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 653A..... 674

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 920C..G 901

>[AY531892.1](#) *Beauveria bassiana* translation elongation factor 1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 1048T..... 1069

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1315C..G 1296

>[OQ352168.1](#) *Simplicillium* sp. SQ-2023a isolate AFMCCC 16b translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 295A..... 316

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 562C..G 543

>[JN715652.1](#) *Trichoderma samuelsii* strain S42 translation elongation factor 1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 832C..... 853

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1099C..A 1080

>[PV273596.1](#) *Protocreopsis chlamydospora* strain CBS 144254 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 241A..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 508C..G 489

>[HQ881011.1](#) *Beauveria sungii* strain ARSEF 7281 elongation factor-1 alpha gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 323T..... 344

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 590C..G 571

>[KM652009.1](#) *Ophiocordyceps sinensis* voucher ARSEF 6282 translation elongation factor 1-alpha gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 273C..... 294

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 540C..G 521

>[OQ858769.1](#) *Trichoderma virens* isolate R-SM109-S1-4 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 805C..... 826

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1072C..G 1053

>[KJ634767.1](#) *Trichoderma citrinoviride* isolate 7987 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 877C..... 898

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1144C..G 1125

>[KX713675.1](#) *Ophiocordyceps unilateralis* strain SERI1 translation elongation factor 1 alpha (TEF) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 319C..... 340

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 586C..G 567

>[ON868821.1](#) *Ophiocordyceps nujiangensis* strain YHCC 8894 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 271C..... 292

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 538C..G 519

>[OR066659.1](#) Ophiocordyceps sp. isolate GTIFB01 translation elongation factor 1-alpha (EF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 235C..... 256

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 502C..G 483

>[KF811537.1](#) Neotyphodium sp. FaTG-5 strain NEA19_G2 translation elongation factor 1-alpha (tefA) gene, complete cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 1131T..... 1152

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1398T..A 1379

>[PQ516506.1](#) Pseudodiploospora sp. isolate SHZ-17-1 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 270T..... 291

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 537C..G 518

>[KF747093.1](#) Tolypocladium tropicale strain IQ45 translation elongation factor 1 alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 277C..... 298

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 544C..G 525

>[MH647777.1](#) Trichoderma atroviride isolate HZA2 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 335C..... 356

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 602C..A 583

>[OP792042.1](#) Trichoderma sp. isolate CEP-754 translation elongation factor 1-alpha gene, partial

cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 296C..... 317

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 563C..G 544

>[PP085550.1](#) *Metapochonia simonovicovae* isolate BiMMF299 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 279T..... 300

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 546C..A 527

>[OR548059.1](#) *Trichoderma tomentosum* isolate 153 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 816C..... 837

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1083C..G 1064

>[ON166729.1](#) *Trichoderma harzianum* isolate NECC21253 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 819C..... 840

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1086C..G 1067

>[LC684182.1](#) *Claviceps miscanthicola* TNS:F-96468 TEF gene for translation elongation factor 1-alpha, partial sequence

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 326T..... 347

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 593T..C 574

>[MH244862.1](#) *Cordyceps fumosorosea* strain AMEF082 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 249C..... 270

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 516C..A 497

>[EU401591.1](#) *Trichoderma longibrachiatum* strain CBS 816.68 translation elongation factor 1 alpha-like (tef1 alpha) gene, partial sequence

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 443C..... 464

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 710C..A 691

>[MW091325.1](#) *Gibellula flava* strain WFS20190625-25 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 295C..... 316

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 562C..G 543

>[AY531948.1](#) *Beauveria bassiana* isolate 7043 translation elongation factor 1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 1083T..... 1104

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1350C..G 1331

>[JN998777.1](#) *Isaria* sp. CNPS0105 translation elongation factor 1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 328T..... 349

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 595C..G 576

>[LC163558.1](#) *Cordyceps brongniartii* EF1-a gene for translation elongation factor 1 alpha, partial cds, isolate: F77

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 326T..... 347

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 593C..G 574

>[OQ927333.1](#) *Beauveria bassiana* strain NRID07 translation elongation factor-1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 268T..... 289

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 535C..G 516

>[JX192796.1](#) *Hevansia novoguineensis* strain EPF020 translation elongation factor 1 alpha (tef) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 295T..... 316

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 562C..G 543

>[OR115176.1](#) *Metarhizium guizhouense* strain YFCC 812 translation elongation factor 1 alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 328T..... 349

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 595T..C 576

>[MK285365.1](#) *Trichoderma* sp. isolate CTCCSJ-F-KZ23687 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 871C..... 892

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1138C..G 1119

>[OR454090.1](#) *Amphichorda monjolensis* isolate COAD 3124 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 244C..... 265

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 511C..G 492

>[OQ440745.1](#) *Ophiocordyceps thanathonensis* voucher HKAS 126056 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 285T..... 306

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 552T..G 533

>[KJ665380.1](#) *Trichoderma atrobrunneum* strain S321 translation elongation factor 1-alpha (tef1)

gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 816C..... 837

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1083C..G 1064

>[HQ881015.1](#) *Beauveria malawiensis* strain ARSEF 4755 elongation factor-1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 323T..... 344

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 590C..G 571

>[KR911897.1](#) *Trichoderma afroharzianum* strain Z19 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 830C..... 851

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1097C..G 1078

>[EU392690.1](#) *Moelleriella zhongdongii* strain P.C. 549 translation elongation factor 1 alpha (TEF1alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 290T..... 311

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 557T..G 538

>[KF811559.1](#) *Neotyphodium* sp. FaTG-2 strain NEA17_G1 translation elongation factor 1-alpha (tefA) gene, complete cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 1113T..... 1134

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1380T..A 1361

>[OQ471220.1](#) *Trichothecium hongkongense* isolate CBS117586 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 241T..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 508C..A 489

>[OR115218.1](#) *Keithomyces* sp. QZW-2023b strain YFCC 849.01 translation elongation factor 1 alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 328C..... 349

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 595T..C 576

>[OR066657.1](#) *Ophiocordyceps* sp. isolate PXIFB01 translation elongation factor 1-alpha (EF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 235C..... 256

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 502C..G 483

>[FJ229490.1](#) *Beauveria* sp. KVL_03_136 elongation factor-1 alpha gene, partial sequence

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 1060T..... 1081

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1327C..G 1308

>[EU248864.1](#) *Metarhizium lepidiotae* culture-collection ARSEF:7412 translation elongation factor 1 alpha (TEF) gene, partial sequence

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 1030T..... 1051

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1297T..C 1278

>[OU720064.1](#) *Claviceps sulcata* partial tef1a gene for translation elongation factor 1 alpha, strain CCC1252

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 325C..... 346

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 592T..C 573

>[OQ124001.1](#) *Trichoderma guizhouense* strain JZB3360017 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 797C..... 818

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1064C..G 1045

>[KJ665570.1](#) *Trichoderma mediterraneum* strain S239 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 821T..... 842

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1088C..G 1069

>[JN715623.1](#) *Trichoderma caerulescens* strain S203 translation elongation factor 1 alpha gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 821C..... 842

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1088C..G 1069

>[EU369018.1](#) *Torrubiella arachnophilus* strain NHJ 10808 translation elongation factor 1 alpha (TEF) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 326C..... 347

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 593C..G 574

>[DQ676607.1](#) *Stachybotrys kampalensis* strain CBS 388.73 translation elongation factor 1 alpha (tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 295C..... 316

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 562C..G 543

>[PV126207.1](#) *Metarhizium brunneum* voucher CHE-CNRCB 540 translation elongation factor (TEF1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 1071T..... 1092

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1338T..C 1319

>[OL322032.1](#) *Ascopolyporus purpuratus* strain BCC88430 translation elongation factor 1-alpha

gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 289C..... 310

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 556C..G 537

>[AF534595.1](#) *Trichoderma strictipile* strain GJS 89-114 translation elongation factor 1-alpha (ef1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 183C..... 204

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 450C..G 431

>[PP464675.1](#) *Niveomyces* sp. strain CGMCC3.25672 translation elongation factor 1-alpha (Tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 298T..... 319

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 565C..G 546

>[PV273632.1](#) *Hydropisphaera fungicola* strain CBS 123912 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 241C..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 508C..G 489

>[PV166496.1](#) *Gibellula penicillioides* strain RCEF7678 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 255A..... 276

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 522C..A 503

>[AY883710.1](#) *Beauveria bassiana* strain ARSEF 7270 translation elongation factor 1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 980T..... 1001

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 1247C..G 1228

>[OQ858736.1](#) *Trichoderma guizhouense* isolate AR68-S3-3 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 813C..... 834

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 1080C..G 1061

>[PV167697.1](#) *Ophiocordyceps pseudobifertilis* voucher GMB 3114 translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 328C..... 349

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 595C..G 576

>[PP860723.1](#) *Trichoderma virens* isolate TrVi15 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 793C..... 814

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 1060C..G 1041

>[JN715620.1](#) *Trichoderma caerulescens* strain S78 translation elongation factor 1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 821C..... 842

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 1088C..G 1069

>[PV755812.1](#) *Trichoderma* sp. strain CFCC 53259A translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 807T..... 828

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 1074C..G 1055

>[GU585569.1](#) *Beauveria bassiana* strain RCEF3506 translation elongation factor 1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 54T..... 75

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 321C..G 302

>[MZ552294.1](#) *Trichoderma virens* strain T10#1-6 translation elongation factor 1 alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 302C..... 323

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 569C..G 550

>[FJ660614.1](#) *Epichloe festucae* strain E2368 translation elongation factor 1-alpha (tefA) gene, complete cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 1135T..... 1156

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1402T..A 1383

>[ON567195.1](#) *Trichoderma ganodermatigerum* voucher CCMJ5245 translation elongation factor EF-1 alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 827C..... 848

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1094C..A 1075

>[KX344433.1](#) *Trichoderma verticillatum* strain HMAS 248742 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 826C..... 847

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1093C..G 1074

>[PQ650481.1](#) *Clonostachys yadongensis* strain HSC1025 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 276T..... 297

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 543C..G 524

>[OQ621373.1](#) *Beauveria bassiana* isolate LPSc1441 translation elongation factor 1-alpha

(EF1alpha) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 259C..... 280
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 526C..G 507

>[KJ665470.1](#) *Trichoderma dorotheae* strain S480 voucher WU:33393 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 798C..... 819
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1065C..G 1046

>[OP797912.1](#) *Aschersonia mackerrasiae* strain BRIP 49212a translation elongation factor 1-alpha gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 320C..... 341
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 587T..G 568

>[XM_056193486.1](#) *Akanthomyces muscarius* uncharacterized protein (LMH87_002091), partial mRNA

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 625T..... 646
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 892C..G 873

>[OU720061.1](#) *Claviceps* sp. KP-2018d partial tef1a gene for translation elongation factor 1 alpha, strain CCC1458

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 325C..... 346
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 592T..C 573

>[KJ665521.1](#) *Trichoderma istrianum* strain S123 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 796C..... 817
 Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 1063C..G 1044

>[KF811544.1](#) *Neotyphodium* sp. FaTG-2 strain NEA32_G2 translation elongation factor 1-alpha (tefA) gene, complete cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 1124T..... 1145

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1391T..A 1372

>[KU529130.1](#) *Trichoderma sinokoningii* strain 8849 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 845C..... 866

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1112C..G 1093

>[LC598958.1](#) *Claviceps humidiphila* TNS:F60506 Ef1-alpha gene for translation elongation factor 1-alpha, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 326C..... 347

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 593T..C 574

>[DQ307527.2](#) *Hypocrea vinosa* strain GJS 99-156 elongation factor 1 alpha (tef1) gene, partial sequence

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 850C..... 871

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1117C..A 1098

>[OQ944549.1](#) *Clonostachys aureofulvella* culture CBS:102838 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 241T..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 508C..A 489

>[AY225858.1](#) *Trichoderma sulphureum* strain GJS 95-190

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 259C..... 280

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 526C..C 507

>[OQ944847.1](#) *Clonostachys solani* f. *nigrovirens* culture CBS:101924 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 241T..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 508C..A 489

>[MZ441385.1](#) *Trichoderma* sp. isolate LZ014 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 808C..... 829

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1075C..G 1056

>[OP797921.1](#) *Aschersonia mackerrasiae* strain BRIP 65580a translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 284C..... 305

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 551T..G 532

>[OP797911.1](#) *Aschersonia mackerrasiae* strain BRIP 62325a translation elongation factor 1-alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 321C..... 342

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 588T..G 569

>[MK524326.1](#) *Trichoderma rossicum* clone MS5 translation elongation factor 1-alpha (EF-1alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 726C..... 747

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 993C..A 974

>[LN810532.1](#) *Acremonium* sp. I AG-2015 partial tef1a gene for translation elongation factor 1

alpha, strain UTHSC 08-2284

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 290C..... 311

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 557C..G 538

>[OQ067542.1](#) *Beauveria bassiana* isolate BCi2Rv19 translation elongation factor 1-alpha (TEF) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 297T..... 318

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 564C..G 545

>[JN819014.1](#) *Hirsutella formicarum* isolate Co23 translation elongation factor 1 alpha (tef) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 328C..... 349

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 595C..G 576

>[PV574443.1](#) *Keithomyces apolzaniae* strain MST-F28739 translation elongation factor 1-alpha (tef) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 293A..... 314

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 560C..A 541

>[DQ069993.1](#) *Aschersonia aleyrodis* strain PC321 elongation factor 1-alpha gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 186T..... 207

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 453T..G 434

>[OQ944824.1](#) *Clonostachys samuelsii* culture CBS:701.97 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268
 Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 241T..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 508C..A 489

>[OQ559121.1](#) *Trichoderma variegatum* strain YMF 1.7533 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 752T..... 773

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1019C..G 1000

>[PV273651.1](#) *Paracylindrocarpon aloicola* strain CBS 126107 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 241C..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 508C..A 489

>[MH397454.1](#) *Claviceps purpurea* voucher DAOMC:250814 translation elongation factor 1-alpha (Ef1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 291C..... 312

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 558T..C 539

>[OQ858722.1](#) *Trichoderma* aff. *harzianum* isolate VO78-S5-2 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 797C..... 818

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1064C..G 1045

>[EF469071.1](#) *Shimizuomyces paradoxus* strain EFCC 6279 translation elongation factor 1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 328C..... 349

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 595C..A 576

>[OQ944600.1](#) *Clonostachys farinosa* culture CBS:115292 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22

Template 241T..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 508C..A 489

>[AY531938.1](#) *Beauveria bassiana* isolate 4933 translation elongation factor 1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 1048T..... 1069

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1315C..G 1296

>[MF416474.1](#) *Hevansia nelumboides* voucher BCC 2190 translation elongation factor 1 alpha (TEF) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 280T..... 301

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 547C..G 528

>[AF543778.1](#) *Claviceps purpurea* strain AEG 97-2 translation elongation factor 1 alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 328C..... 349

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 595T..C 576

>[OR757314.1](#) *Trichoderma paratroviride* isolate T0023 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 814C..... 835

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 1081C..A 1062

>[OQ470872.1](#) *Emericellopsis glabra* isolate CBS119.40 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
Template 241C..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
Template 508C..G 489

>[MF872613.1](#) *Ophiocordyceps thanathonensis* culture MFLUCC:16-2909 translation elongation

factor 1-alpha (EF1-alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 298T..... 319

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 565T..G 546

>[MG383495.1](#) *Trichoderma* sp. 4 WYZ-2018 strain HMAS 275664 translation elongation factor 1-alpha (TEF1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 879C..... 900

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1146C..G 1127

>[OP562790.1](#) *Trichoderma atrobrunneum* isolate HF3S32 elongation factor 1-alpha (tef-1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 847C..... 868

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1114C..G 1095

>[OQ944879.1](#) *Clonostachys zelandiae-novae* culture CBS:100978 translation elongation factor 1-alpha (tef1) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 241T..... 262

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 508C..A 489

>[JN715663.1](#) *Trichoderma hispanicum* strain S536a translation elongation factor 1 alpha gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 837C..... 858

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20
 Template 1104C..G 1085

>[MW267395.1](#) *Trichoderma harzianum* isolate TA2-3 translation elongation factor 1-alpha (TEF-1 alpha) gene, partial cds

product length = 268

Forward primer 1 TGGTACAAGGGNTGGGAGAAGG 22
 Template 786C..... 807

Reverse primer 1 GCTGCTCGTGGTGCATNTCN 20

Template 1053C..G 1034

If you want to allow any of the unintended targets, check the box(es) next to the ones you accept and try again to re-search for specific primers

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